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(54) **CHANGING VOLATILITY VIA SELECTABLE
ACTIVE SYMBOL DISPLAY POSITIONS**

(71) Applicant: **IGT**, Las Vegas, NV (US)

(72) Inventors: **Yu Chong**, Reno, NV (US); **Nick Hillstrom**, Reno, NV (US); **Nathan Ajhar**, Sparks, NV (US); **Chris Peterson**, Reno, NV (US)

(73) Assignee: **IGT**, Las Vegas, NV (US)

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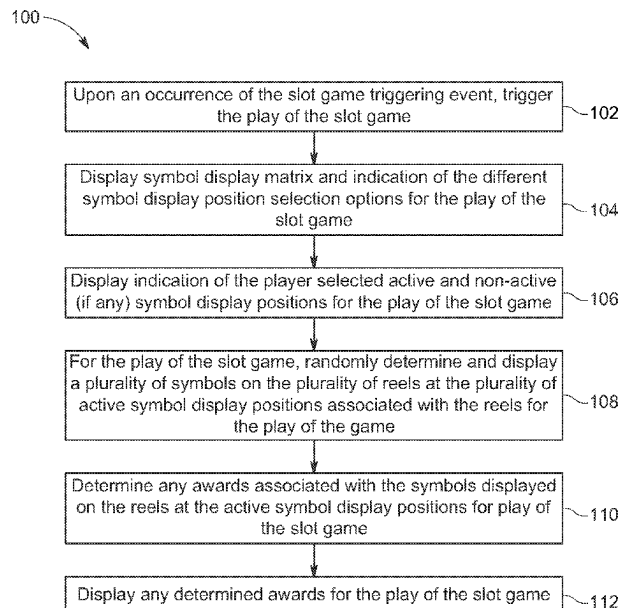
Primary Examiner — Paul A D'Agostino

(74) *Attorney, Agent, or Firm* — Neal, Gerber & Eisenberg LLP

(57) **ABSTRACT**

Gaming systems and methods that provides plays of a game wherein the selection of different symbol display positions of a symbol display position matrix to be active symbol display positions changes volatility of the plays of the game. The systems and methods also provide plays of a game wherein the selection of different symbol display positions of a symbol display position matrix to be non-active symbol display positions changes volatility of the plays of the game.

17 Claims, 14 Drawing Sheets



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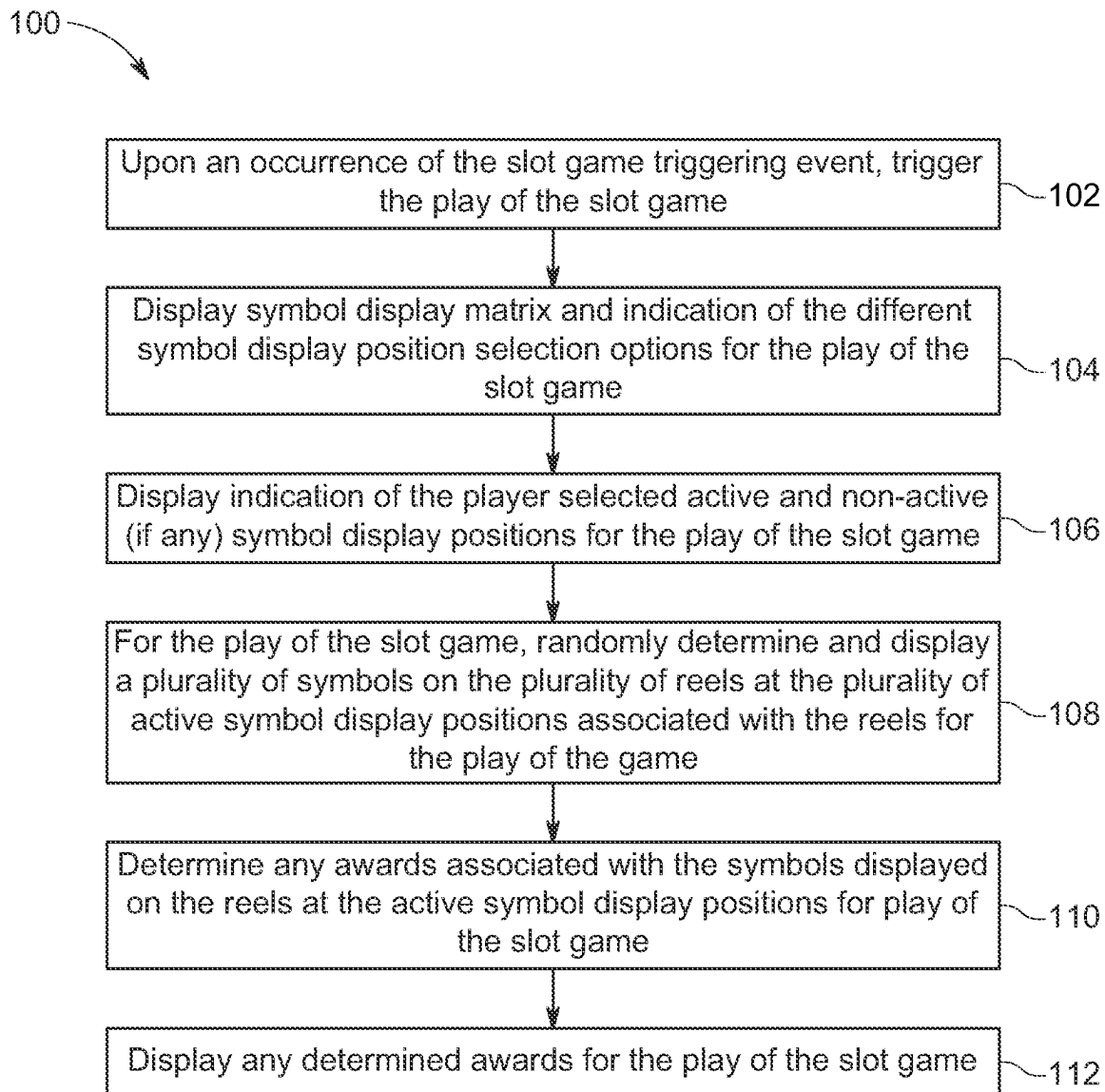


FIG. 1

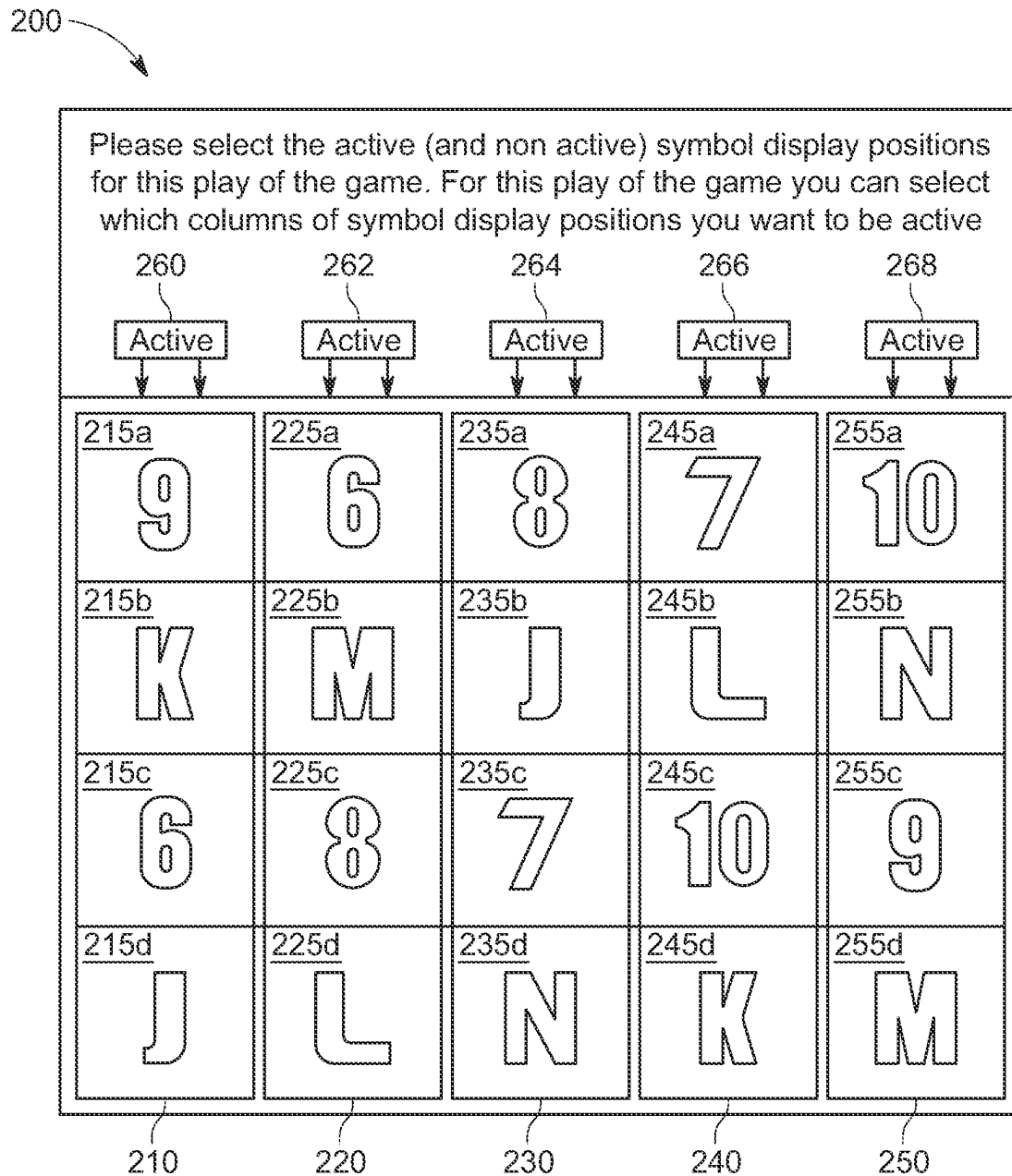


FIG. 2A

200

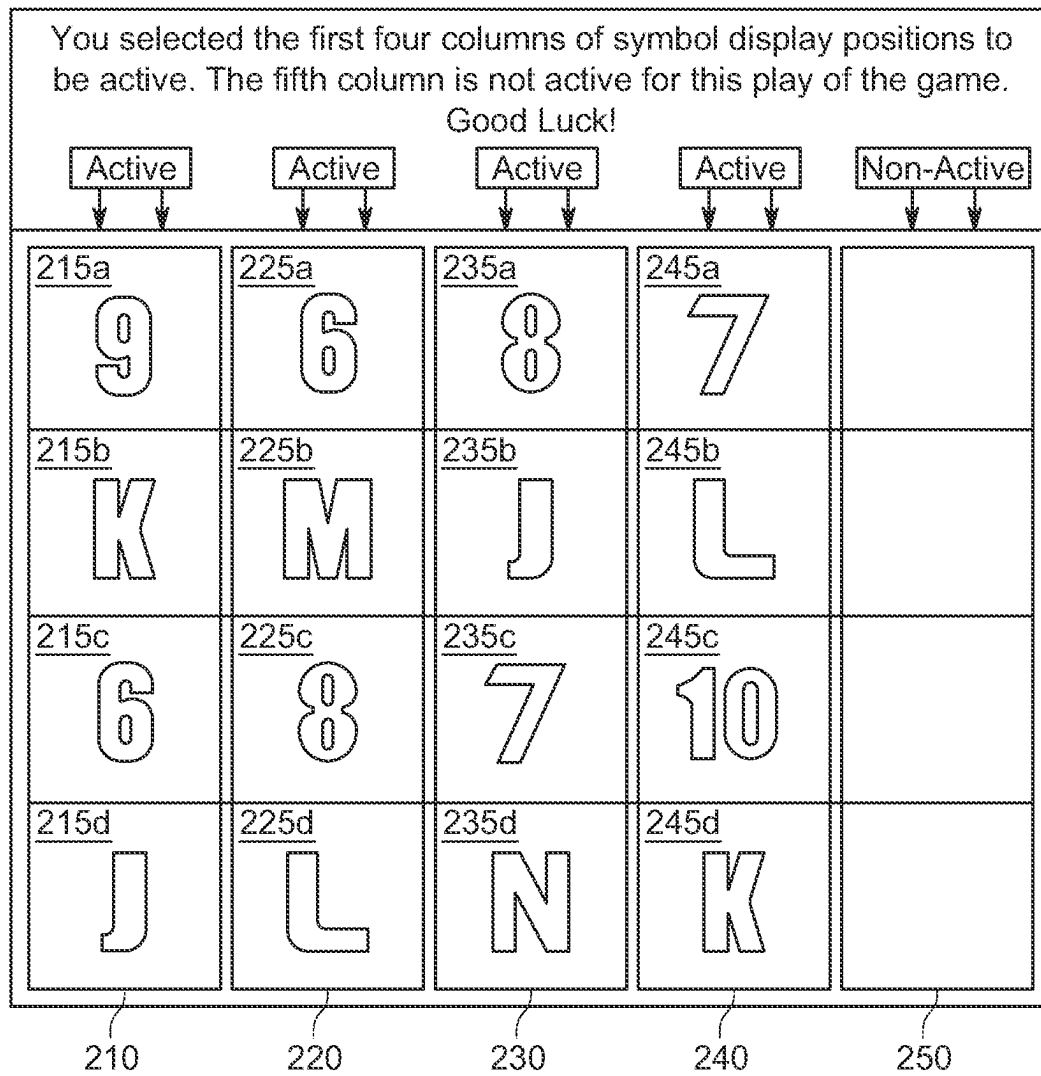


FIG. 2B

200

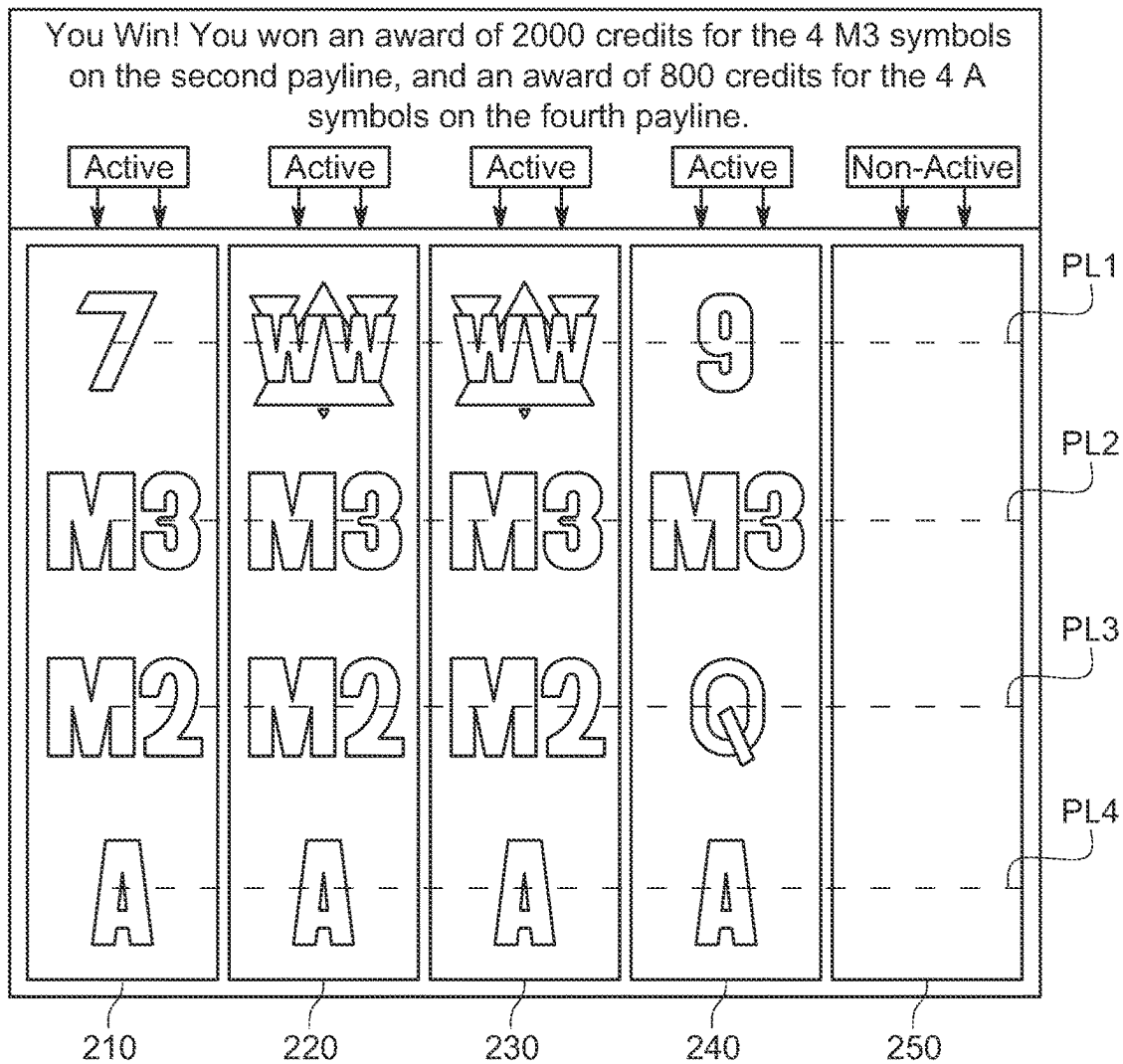


FIG. 2C

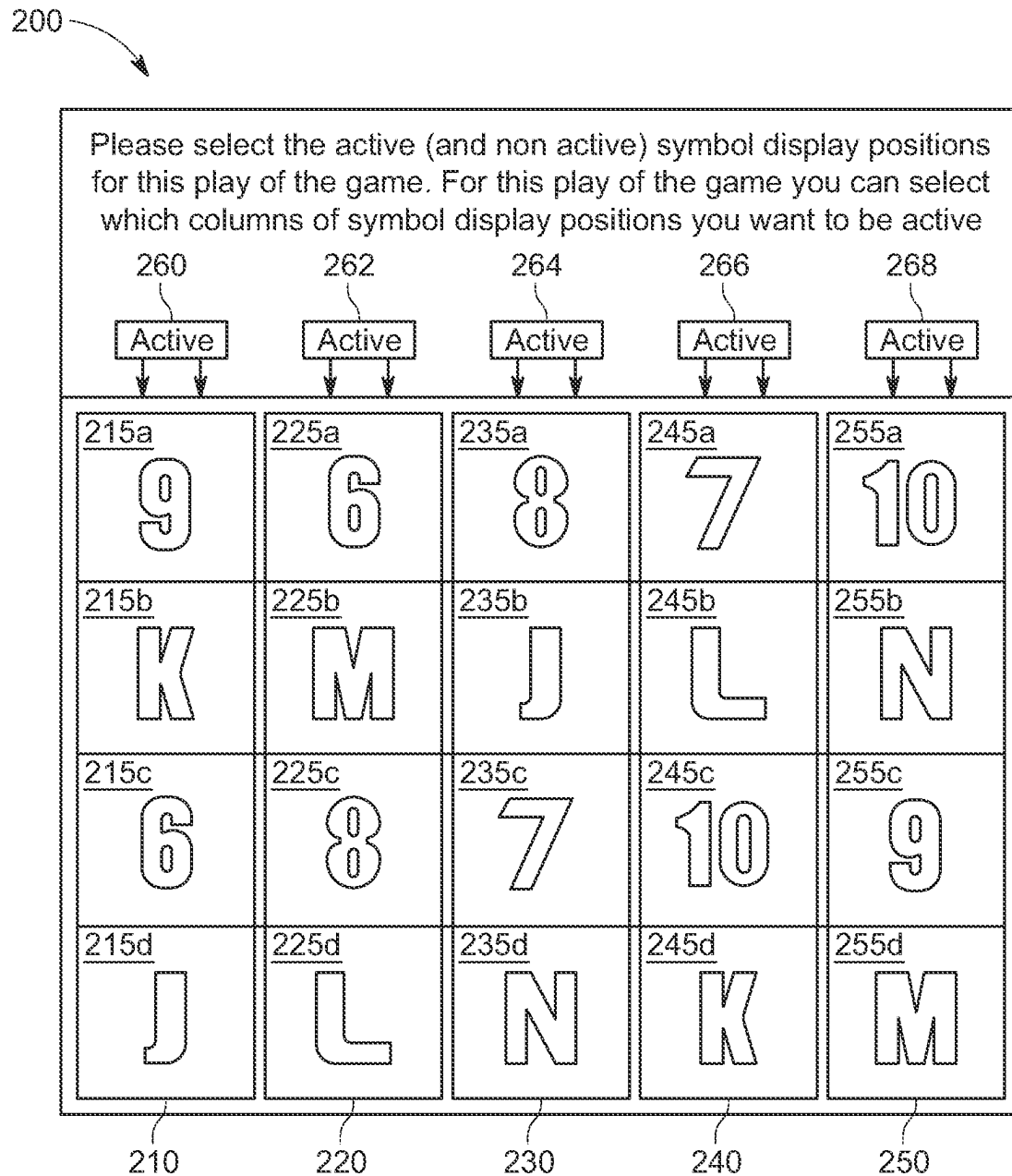


FIG. 3A

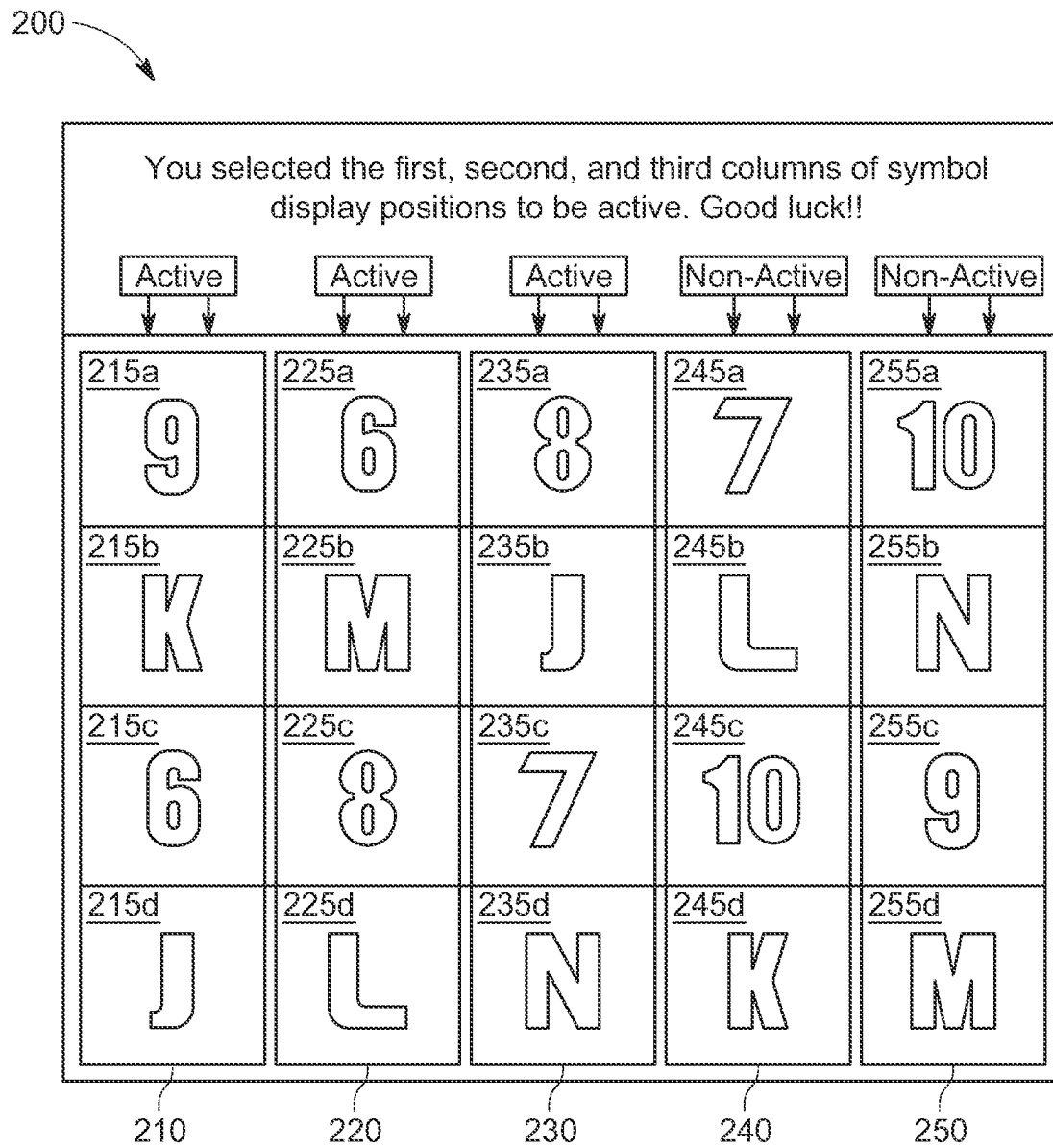


FIG. 3B

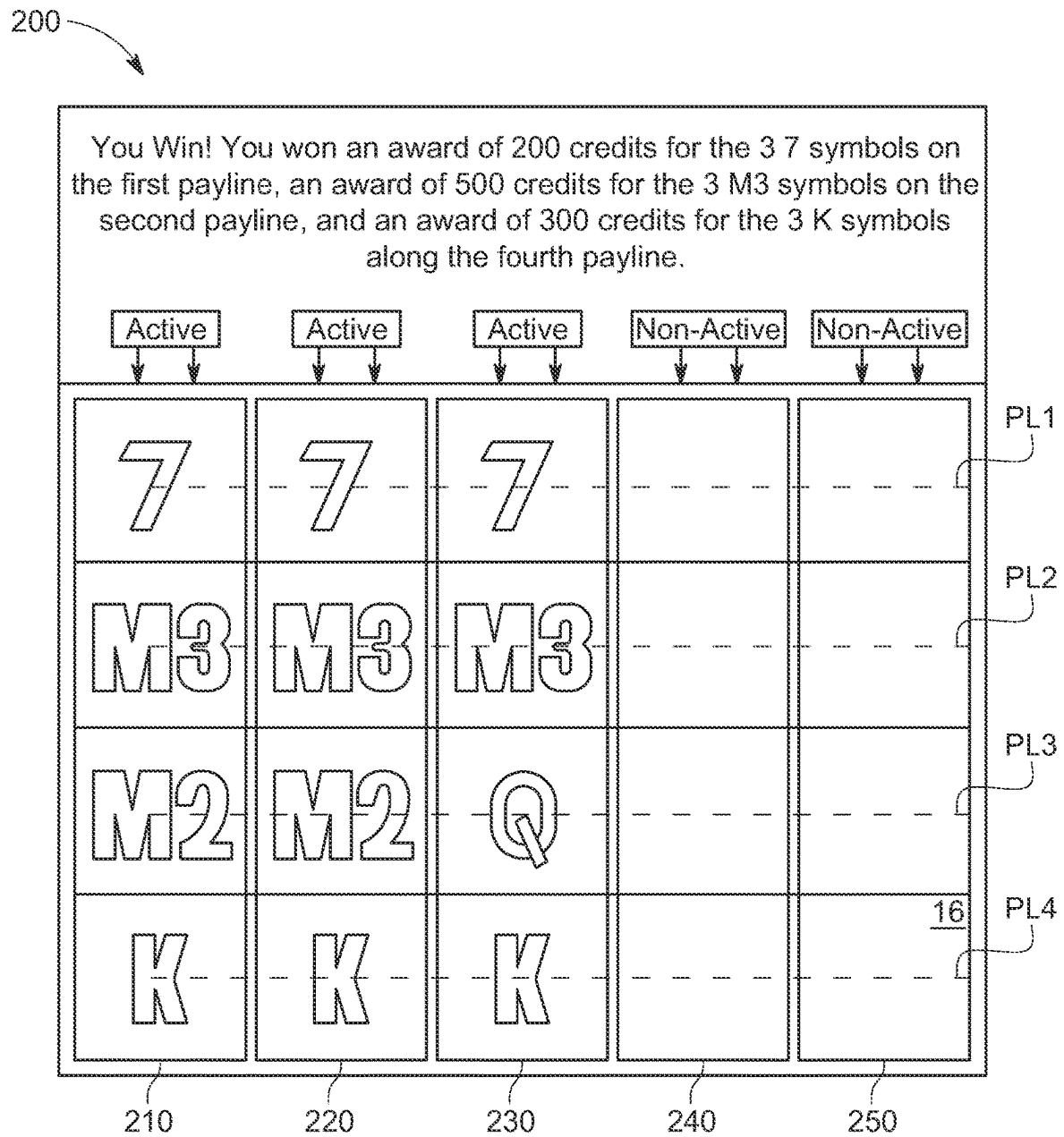


FIG. 3C

200

Please select the active (and non-active) symbol display positions for the play of the game. For this play of the game you can select which rows of symbol display positions you want to be active.

215a 9	225a 6	235a 8	245a 7	255a 10	270 Active Non-Active
215b K	225b M	235b J	245b L	255b N	272 Active Non-Active
215c 6	225c 8	235c 7	245c 10	255c 9	274 Active Non-Active
215d J	225d L	235d N	245d K	255d M	276 Active Non-Active
210	220	230	240	250	

FIG. 4A

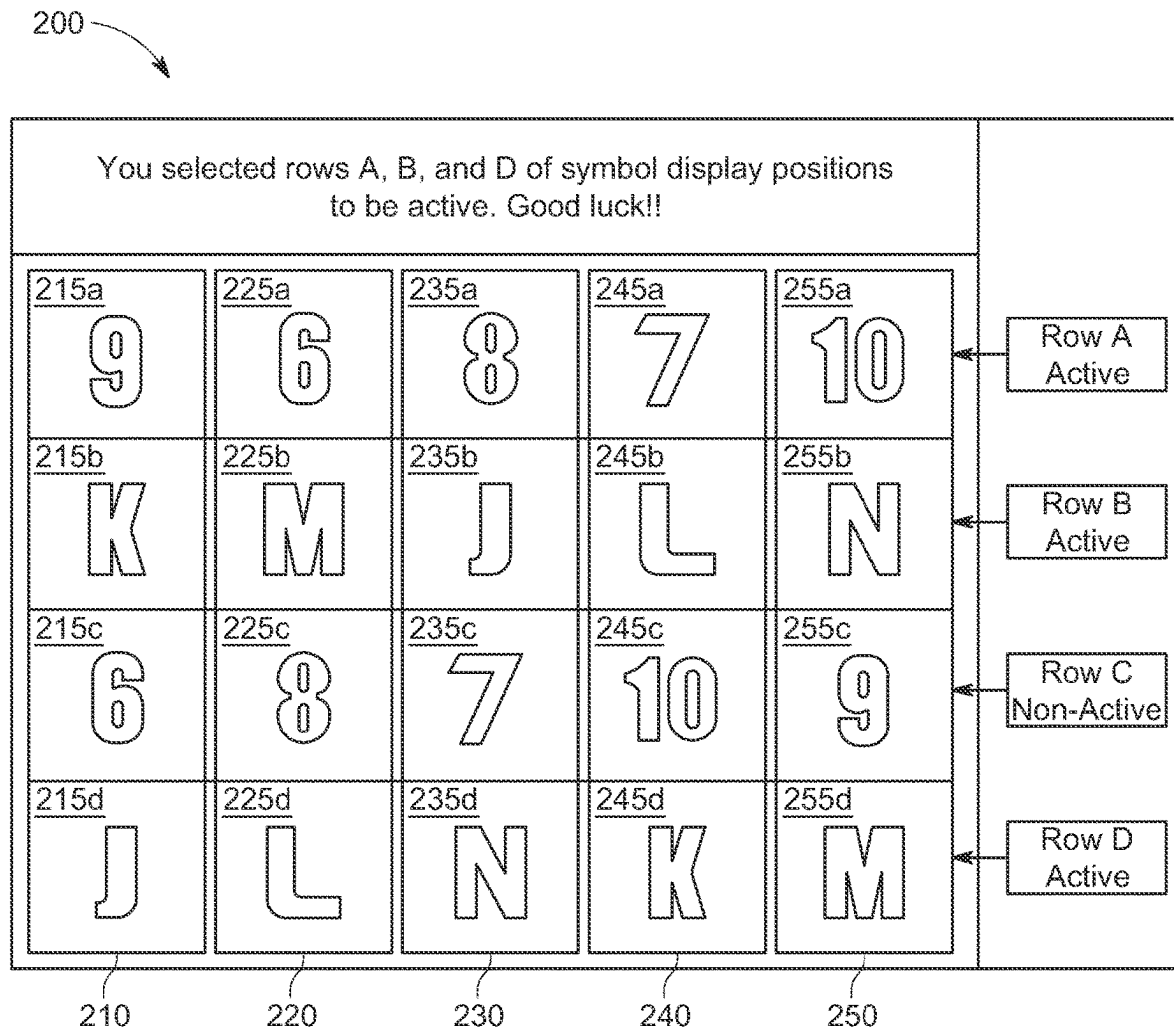


FIG. 4B

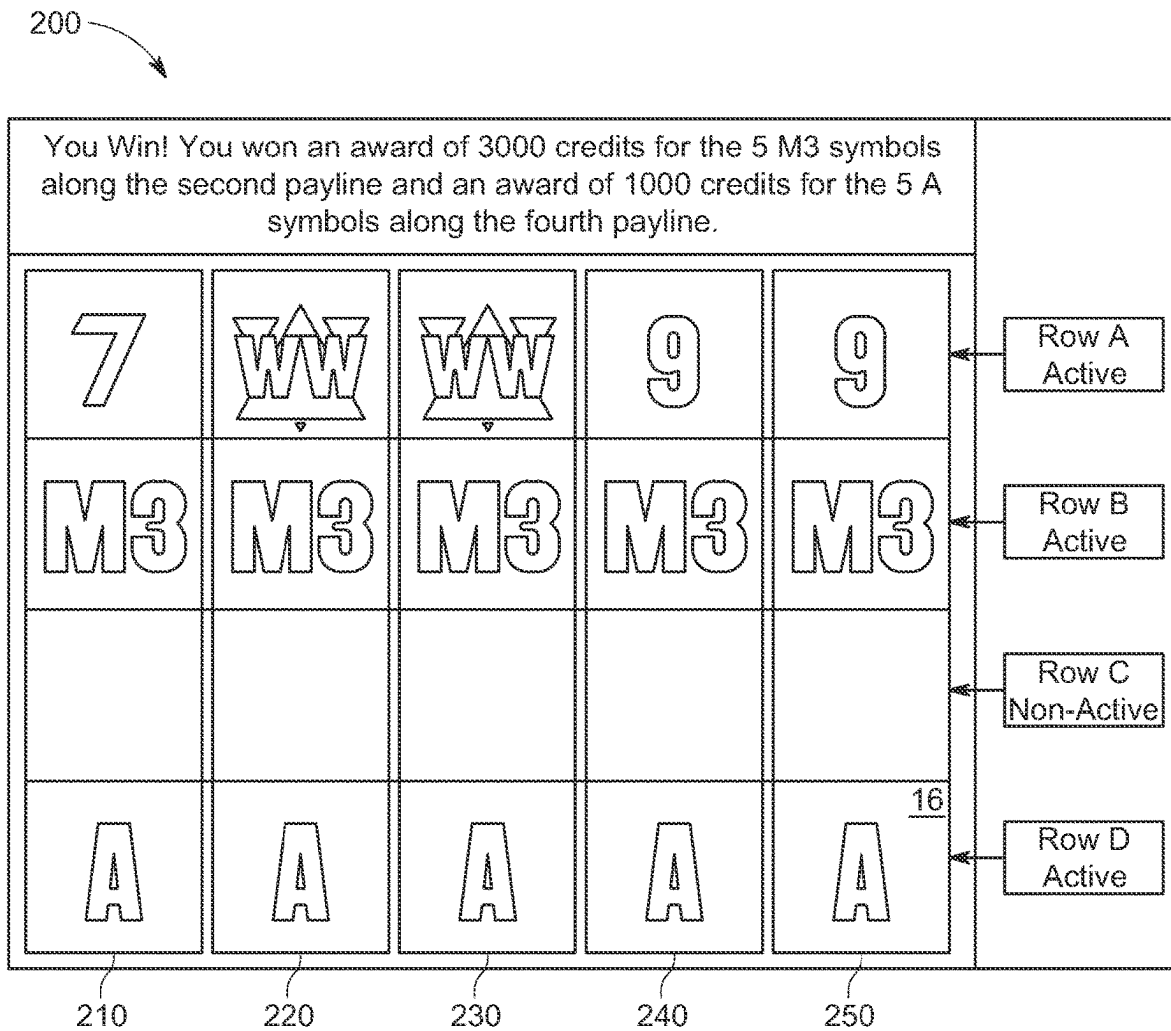


FIG. 4C

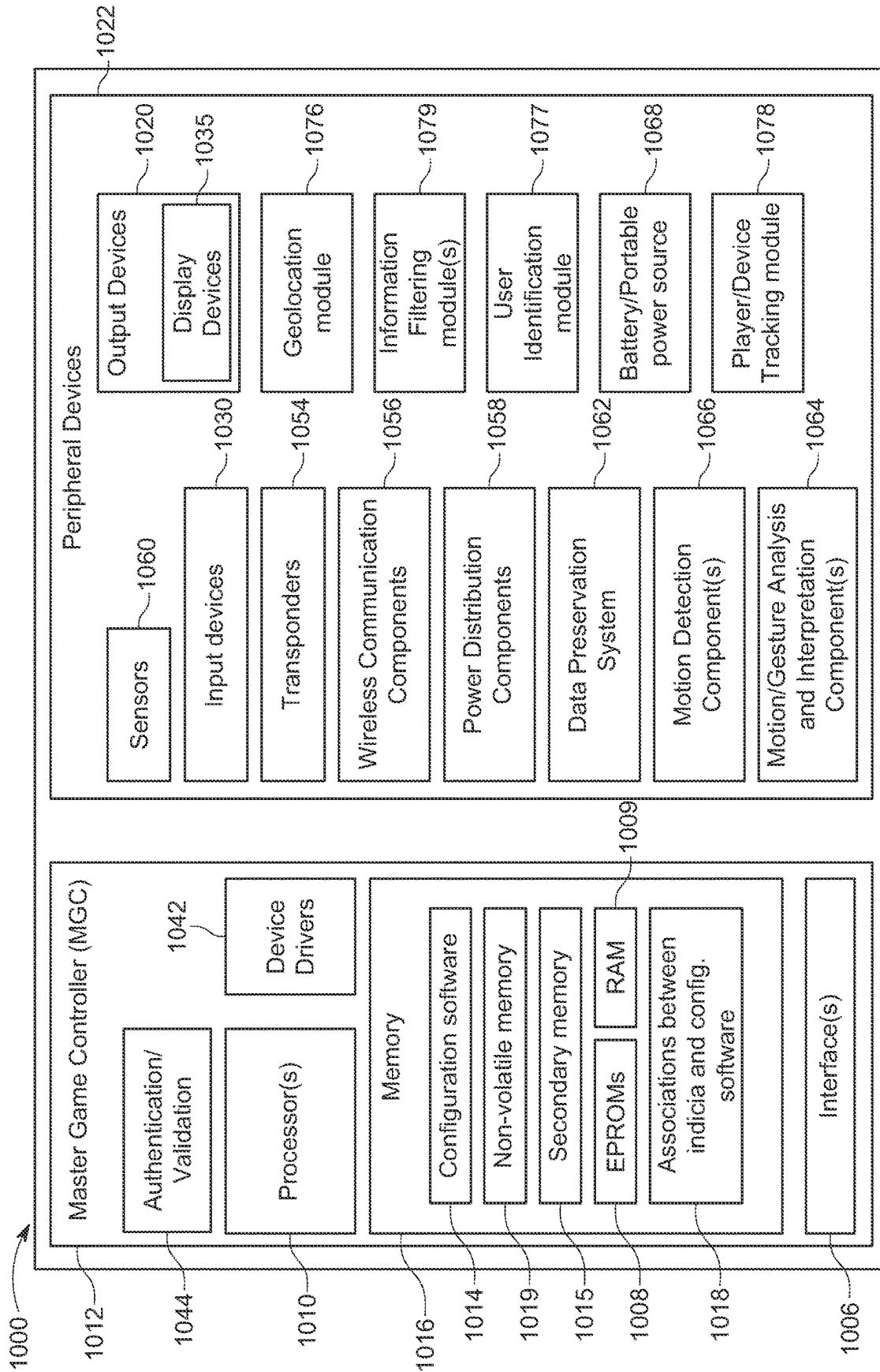


FIG. 5

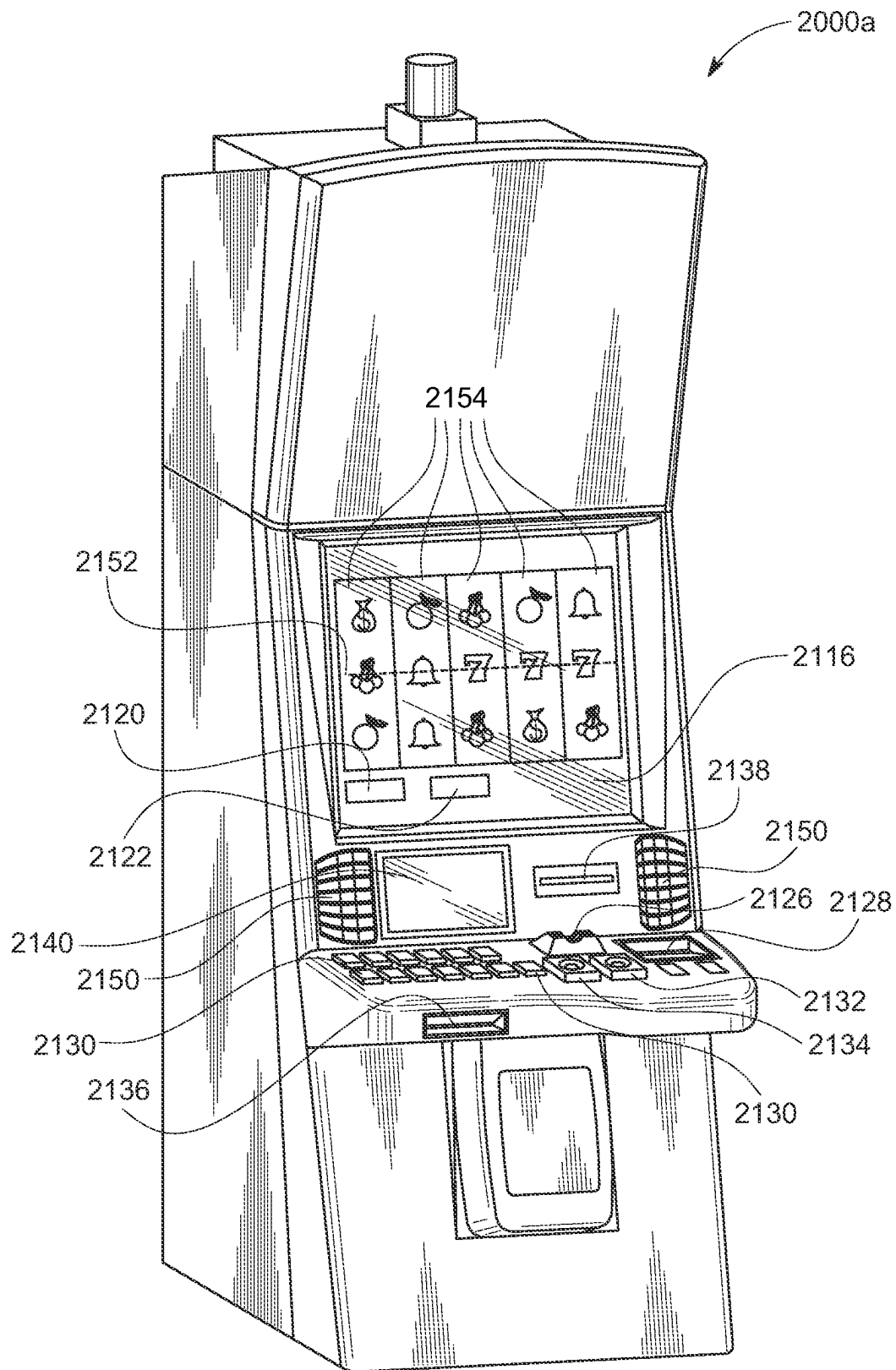


FIG. 6A

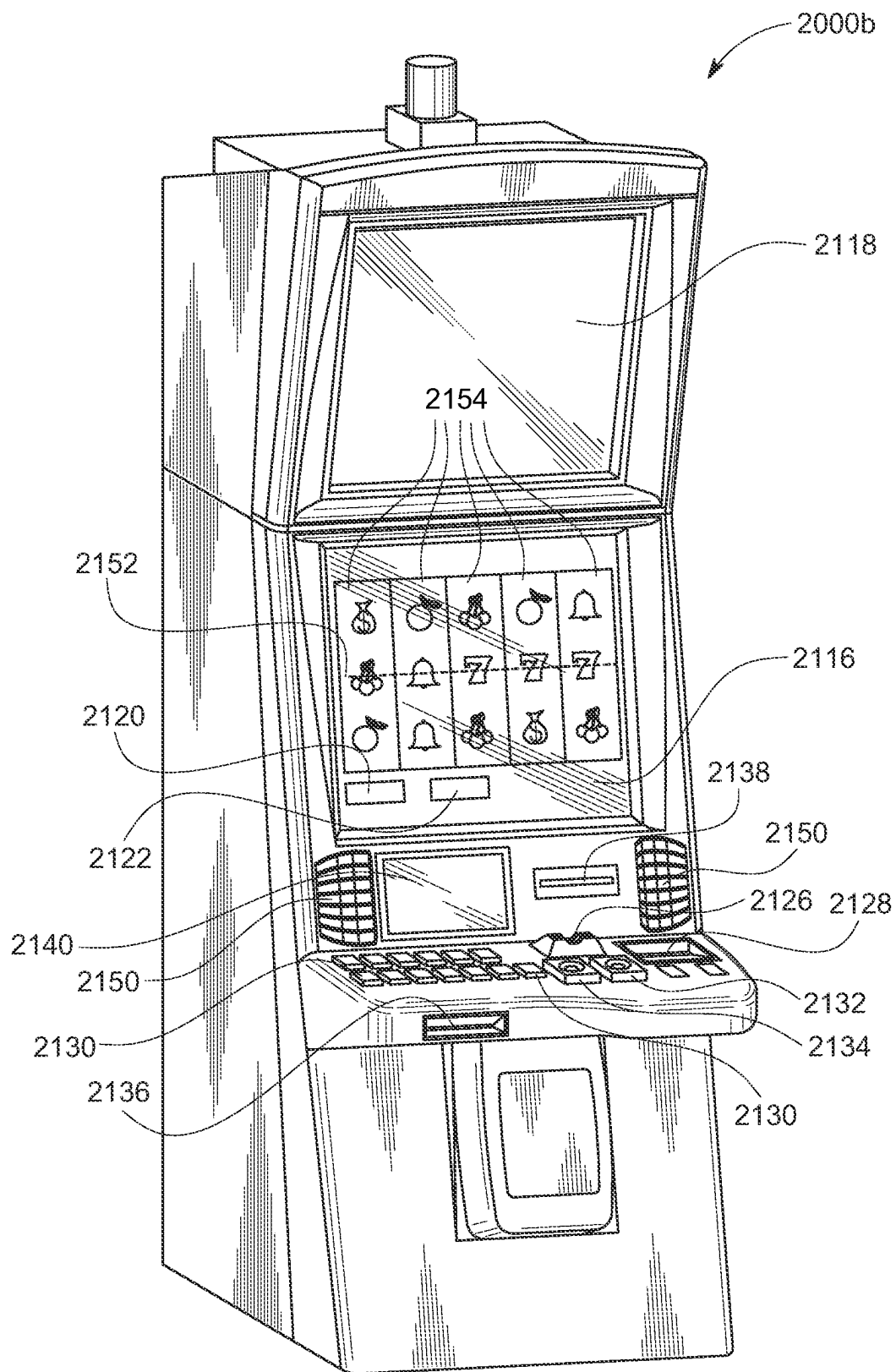


FIG. 6B

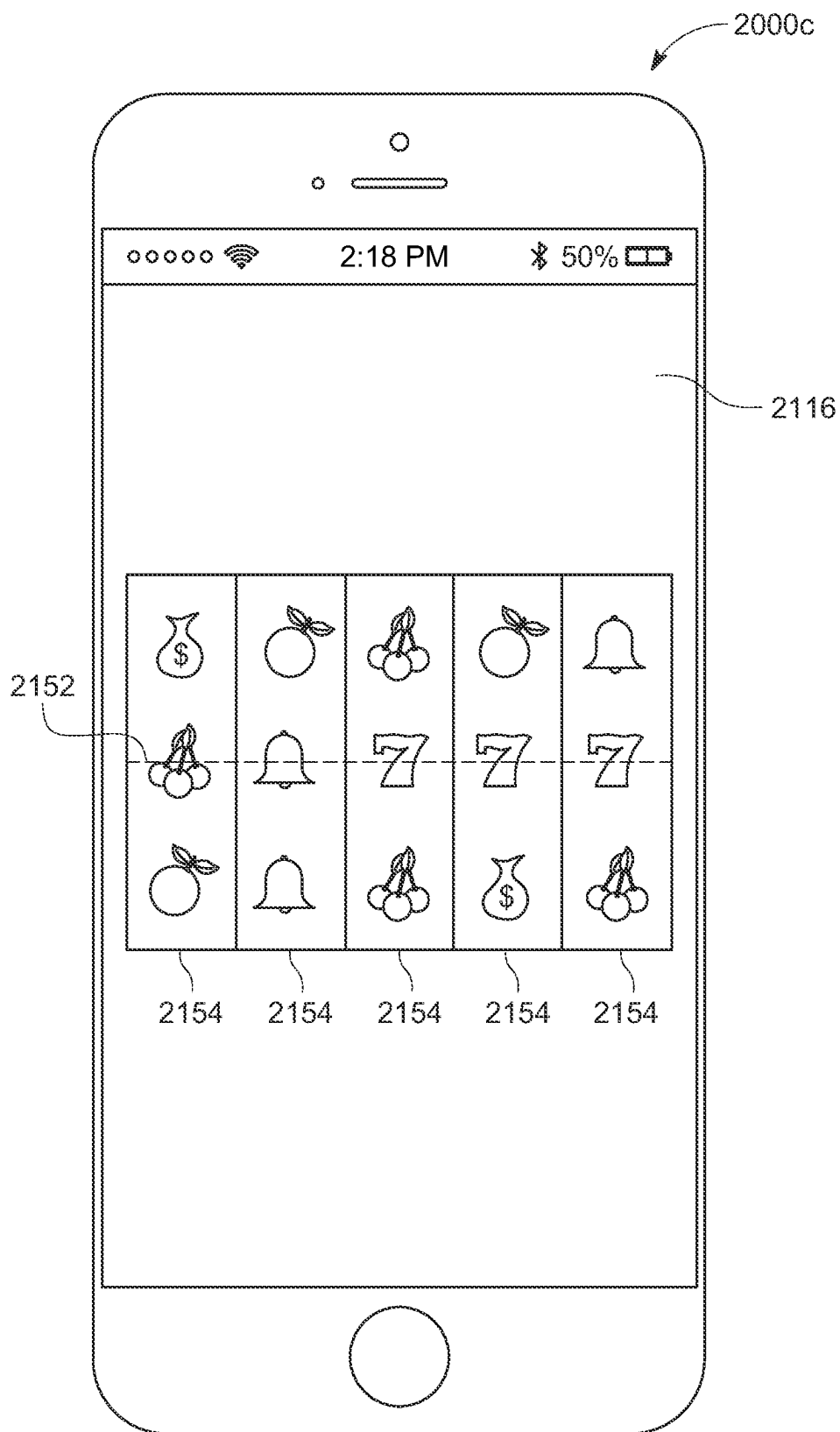


FIG. 6C

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CHANGING VOLATILITY VIA SELECTABLE ACTIVE SYMBOL DISPLAY POSITIONS

BACKGROUND

The present disclosure relates to changing volatility via selectable active symbol display positions.

Gaming machines may provide players awards in primary games. Gaming machines may require the player to place a wager to activate the primary game. The award may be based on the player obtaining a winning symbol or symbol combination and on the amount of the wager.

BRIEF SUMMARY

In various embodiments, the present disclosure relates to a gaming system including a processor and a memory device that stores a plurality of instructions that, when executed by the processor in association with a play of a game, cause the processor to: cause a display, by a display device, of symbol displays including different symbols; cause a display, by the display device, of an indication of selectable symbol display positions associated with the symbol displays; and cause a display, by the display device, of an indication of a selected plurality of the symbol display positions as being active symbol display positions for the play of the game. The plurality of instructions, when executed by the processor in association with the play of the game, further cause the processor to: cause a display, by the display device, of a plurality of the different symbols on the symbol displays displayed at the plurality of active symbol display positions; and cause a display, by the display device, of an indication of a first quantity of same symbols at a plurality of the active symbol display positions, wherein whether the first quantity of same symbols is a winning symbol combination depends on the symbol display positions selected as the active symbol display positions, wherein if a first quantity of symbol display positions are active symbol display positions, the first quantity of same symbols at the active symbol display positions forms the winning symbol combination, and wherein if a greater second quantity of symbol display positions are active symbol display positions, the first quantity of same symbols at the active symbol display positions does not form the winning symbol combination.

In various other embodiments, the present disclosure relates to a gaming system including a processor and a memory device that stores a plurality of instructions that, when executed by the processor, cause the processor to: cause a display, by a display device, of symbol displays including different symbols; and cause a display, by the display device, of an indication of selectable symbol display positions associated with the symbol display. The plurality of instructions, when executed by the processor, further cause the processor to, for a first play of a game: cause a display, by the display device, of an indication of all of the symbol display positions as being first active symbol display positions for the first play of the game; cause a display, by the display device, of a first plurality of the different symbols on the symbol displays displayed at the first active symbol display positions; and cause a display, by the display device, of an indication of a first quantity of same symbols at the first active symbol display positions, wherein whether the first quantity of same symbols is a winning symbol combination depends on the symbol display positions selected as the first active symbol display positions, wherein the first quantity of same symbols at the first active symbol display positions does not form the winning symbol combination. The plu-

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ality of instructions, when executed by the processor, further cause the processor to, for a second play of the game: cause a display, by the display device, of an indication of less than all of the symbol display positions as being second active symbol display positions for the second play of the game; cause a display, by the display device, of a second plurality of the different symbols on the symbol displays displayed at the second active symbol display positions; and cause a display, by the display device, of an indication the first quantity of same symbols at the second active symbol display positions, wherein whether the first quantity of same symbols is the winning symbol combination depends on the symbol display positions selected as the second active symbol display positions, wherein the first quantity of same symbols at the second active symbol display positions form the winning symbol combination.

In various other embodiments, the present disclosure relates to a gaming system including a processor and a memory device that stores a plurality of instructions that, when executed by the processor, cause the processor to: cause a display, by a display device, of symbol displays including different symbols; and cause a display, by the display device, of an indication of selectable symbol display positions associated with the symbol display. The plurality of instructions, when executed by the processor, further cause the processor to, for a first play of a game: cause a display, by the display device, of an indication of a first quantity of the symbol display positions as being first active symbol display positions for the first play of the game; cause a display, by the display device, of a first plurality of the different symbols on the symbol displays displayed at the first active symbol display positions; and cause a display, by the display device, of an indication of a first quantity of same symbols at the first active symbol display positions, wherein whether the first quantity of same symbols is a winning symbol combination depends on the symbol display positions selected as the first active symbol display positions, wherein the first quantity of same symbols at the first active symbol display positions does not form the winning symbol combination. The plurality of instructions, when executed by the processor, further cause the processor to, for a second play of the game: cause a display, by the display device, of an indication of a second quantity of the symbol display positions as being second active symbol display positions for the second play of the game; cause a display, by the display device, of a second plurality of the different symbols on the symbol displays displayed at the second active symbol display positions; and cause a display, by the display device, of an indication the first quantity of same symbols at the second active symbol display positions, wherein whether the first quantity of same symbols is the winning symbol combination depends on the symbol display positions selected as the second active symbol display positions, wherein the first quantity of same symbols at the second active symbol display positions form the winning symbol combination.

Additional features are described herein and will be apparent from the following Detailed Description and the figures.

BRIEF DESCRIPTION OF SEVERAL VIEWS OF THE DRAWINGS

FIG. 1 is a flow chart of an example process for operating a gaming system that enables the selection of symbol display positions of a symbol display matrix to be active symbol

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display positions, wherein the selection of different symbol display positions to be active symbol display positions changes volatility.

FIGS. 2A, 2B, and 2C are front views of example screen displays of an example play of a game that includes the selection of symbol display positions of a symbol display position matrix to be active symbol display positions and that determines volatility of the play of the game based on such selections in accordance with one example embodiment of the present disclosure.

FIGS. 3A, 3B, and 3C are front views of example screen displays of an example play of a game that includes the selection of symbol display positions of a symbol display position matrix to be active symbol display positions and that determines volatility of the play of the game based on such selections in accordance with one example embodiment of the present disclosure.

FIGS. 4A, 4B, and 4C are front views of example screen displays of an example play of a game that includes the selection of symbol display positions of a symbol display position matrix to be active symbol display positions and that determines volatility of the play of the game based on such selections in accordance with one example embodiment of the present disclosure.

FIG. 5 is a schematic block diagram of one embodiment of an electronic configuration of an example gaming system disclosed herein.

FIGS. 6A and 6B are perspective views of example alternative embodiments of the gaming system disclosed herein.

FIG. 6C is a front view of an example personal gaming device of the gaming system disclosed herein.

DETAILED DESCRIPTION

In various embodiments, the present disclosure relates to systems and methods that provide plays of a game wherein the selection of different symbol display positions of a symbol display position matrix to be active symbol display positions changes volatility of the plays of the game. The present disclosure thus also relates to systems and methods that provide plays of a game wherein the selection of different symbol display positions of a symbol display position matrix to be non-active symbol display positions changes volatility of the plays of the game. It should thus be appreciated that the present disclosure can be implemented either via: (1) the selection of active symbol display positions (that indirectly result in the selection of the non-active symbol display positions); or (2) the selection of non-active symbols display positions (that indirectly result in the selection of the active symbol display positions). In various embodiments, the symbol display positions of the symbol display matrix that are not selected to be the active symbol display positions are the non-active symbol display positions. In other words, in various embodiments, the selection of the active symbol display positions is via the selection of the non-active symbol display positions (i.e., the symbol display positions of the symbol display matrix that are selected to be the non-active symbol display positions determines the active symbol display positions).

In various embodiments, the gaming system employs a plurality of symbol displays (such as reels) and a plurality of symbol display positions associated with the symbol displays (such as the reels). The plurality of symbol display positions form the symbol display matrix. For each of one or more plays of a game, one or more of the symbol display positions can be selected to be active for that play of the

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game. Thus, for each of one or more plays of a game, one or more of the symbol display positions can be selected to be non-active for that play of the game. In various embodiments, the gaming system enables the player to select the active symbol display positions (and thus also indirectly the non-active symbol display positions). In various other embodiments, the gaming system enables the player to select the non-active symbol display positions (and thus also indirectly the active symbol display positions).

In various embodiments, the gaming system enables one or more of the symbol display positions to be individually selected to be active for a play of a game. In various embodiments, the gaming system enables one or more of the symbol display positions to be individually selected to be non-active for a play of a game.

In various embodiments, the gaming system enables one or more groups of symbols displays to be selected to be active for a play of a game. In various embodiments, the gaming system enables one or more groups of symbols displays to be selected to be non-active for a play of a game. In various embodiments, each group is one or more columns of the symbol display positions. In various embodiments, each group is part of one or more columns of the symbol display positions. In various such embodiments, each group is one or more rows of the symbol display positions. In various such embodiments, each group is part of one or more rows of the symbol display positions. In various such embodiments, each group is other than a row or column of symbol display positions.

In various embodiments, the gaming system enables a player to directly select the symbol display positions that are active for each of one or more plays of a game. In various embodiments, the gaming system enables a player to directly select the symbol display positions that are non-active for each of one or more plays of a game.

In various embodiments, the gaming system directly selects the symbol display positions that are active for each of one or more plays of a game. In various embodiments, the gaming system directly selects the symbol display positions that are non-active for each of one or more plays of a game. In various embodiments, the gaming system randomly selects the symbol display positions that are active for each of one or more plays of a game. In various embodiments, the gaming system randomly selects the symbol display positions that are non-active for each of one or more plays of a game.

In the various example embodiments described herein, the symbol display positions associated with symbol display such as the reels can be selected to be active or non-active. In various such embodiments, this can be done by changing the size of such symbol displays (such as by changing the size of such video reels). In various such embodiments, this can be done by otherwise changing the symbol displays (such as by otherwise changing the video reels).

In various embodiments, responsive to an occurrence of a symbol display position selection event for a play of a game, the gaming system enables or causes the selection of any active symbol display positions for that play of the game. In various embodiments, the symbol display position selection event for a play of a game is a wager made on the play of the game (such as for a primary wagering game). In various embodiments, the symbol display position selection event can occur for every play of a game. In various embodiments, the symbol display position selection event can occur for only certain plays of a game. In various embodiments, the gaming system can provide the symbol display position selection event based on a predetermined order or in a

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predetermined manner. In various embodiments, the gaming system can provide the symbol display position selection event based on a random determination.

In various embodiments, the gaming system enables the player to select the active (or non-active) symbol display positions for each individual play of a game. In various embodiments, the gaming system enables the player to select the active (or non-active) symbol display positions for more than one individual play of a game at one time. Thus, the gaming system enables the player to change the volatility for one or more plays of a game at one time.

In various embodiments, for each active symbol display position for a play of the game, the symbol display (such as the reel) that is configured to display symbols at that active symbol display position can display a symbol (such as a randomly determined symbol on that reel) at that active symbol display position.

In various embodiments, for each non-active symbol display position for a play of the game, the symbol display (such as the reel) that is configured to display symbols at that non-active symbol display position is partially or fully disabled (and thus does not display any symbol at that non-active symbol display position).

In various embodiments, for each non-active symbol display position for a play of the game, the symbol display (such as the reel) that is configured to display symbols at that non-active symbol display position can display a symbol at that non-active symbol display position, but such symbol is not evaluated for that play of the game.

In various embodiments, the gaming system is configured such that: (1) the lower the quantity of active symbol display positions, the higher the probability of obtaining a winning symbol combination, but the lower the win amount for obtaining that winning symbol combination; and (2) the higher the quantity of active symbol display positions, the lower the probability of obtaining a winning symbol combination, but the higher the win amount for obtaining that winning symbol combination. Thus, as the quantity of active symbol display positions changes, the volatility changes. This is further illustrated by the embodiments and examples explained below.

In various embodiments, the symbol combinations that are winning symbol combinations are based on the quantity of active symbol display positions. Thus, in such embodiments, the volatility changes based on which symbols combinations are winning symbol combinations.

In various example embodiments, before or after the player makes a wager on a play of a game, the gaming system displays to the player an option of how many and/or which columns the player would like to make active for the play of the game. In various embodiments, the fewer the quantity of columns that are active the greater the chance of winning an award. In various such embodiments, the fewer the quantity of columns that are active, the lower the average expected award. In various such embodiments, the gaming system can display indications of such probability and/or award variances (and thus displays indications of the volatility changes).

In various example embodiments, before or after the player makes a wager on a play of a game, the gaming system displays to the player an option of how many and/or which rows the player would like to make active for the play of the game. In various embodiments, the fewer the quantity of rows that are active the greater the chance of winning an award. In various such embodiments, the fewer the quantity of rows that are active, the lower the average expected award. In various such embodiments, the gaming system can

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display indications of such probability and/or award variances (and thus displays indications of the volatility changes).

In one example embodiment, the gaming system enables the player to select three columns (reels) out of a five column (reel) slot game (i.e., the player selects the symbol display positions associated with first three of the reels to be active). The player's chances of obtaining a winning outcome from the play of the game are increased by 20%. In various embodiments, based on the quantity of columns in use, the gaming system changes or employs suitable winning symbol combinations associated with the symbols on the reels to provide a relatively higher probability forming a winning symbol combination. In one such example embodiment, to get a winning symbol combination, the player must obtain as many symbols in a row for as many columns are active (e.g., two matching symbols in a row with two active columns is a winning symbol combination; three matching symbols in a row with three active columns is a winning symbol combination; four matching symbols in a row with four active columns is a winning symbol combination; and five matching symbols in a row with five active columns is a winning symbol combination).

In one such example play of the game, if there are four active symbol display positions along a payline, a four symbol combination displayed at such active symbol display positions is a winning symbol combination, but a three symbol combination is not a winning symbol combination. In this example, for instance, a 7-7-7-7 symbol combination along the payline would be a winning combination, but a 7-7-7 symbol combination along the payline would not be a winning combination. For the same example play of the game, if there are three active symbol display positions along a payline, a three symbol combination displayed at such active symbol display positions is a winning symbol combination, but a two symbol combination is not a winning symbol combination. In this example, for instance, a 7-7-7 symbol combination along the payline would be a winning combination, but a 7-7 symbol combination along the payline would not be a winning combination. Thus, the volatility can be changed for the same play of the game based on the selection of active symbol display positions.

In various other embodiments, based on the quantity of columns in use, the gaming system changes or employs suitable probabilities associated with the symbols on the reels to provide a relatively higher probability of generating three symbols in a row on such active columns to form a winning symbol combination.

In various embodiments, the gaming system is configured such that: (1) the lower the quantity of columns of active symbol display positions, the higher the probability of obtaining a winning symbol combination, but the lower the win amount for obtaining that winning symbol combination; and (2) the higher the quantity of columns of active symbol display positions, the lower the probability of obtaining a winning symbol combination, but the higher the win amount for obtaining that winning symbol combination.

In various embodiments, the gaming system is configured such that: (1) the lower the quantity of rows of active symbol display positions, the higher the probability of obtaining a winning symbol combination, but the lower the win amount for obtaining that winning symbol combination; and (2) the higher the quantity of rows of active symbol display positions, the lower the probability of obtaining a winning symbol combination, but the higher the win amount for obtaining that winning symbol combination.

In various embodiments, a game play triggering event includes the gaming system receiving the player wager for the play of the game, the gaming system receiving the player selection of active symbol display positions of the symbol display matrix for the play of the game. In various embodiments, based on the selected active symbol display positions, if necessary, the gaming system selects the symbol displays, configures the symbol display, or reconfigures the symbol displays associated with the symbol display matrix based on the respective selected active symbol display positions, and displays those symbol displays. In various embodiments, based on the selected active symbol display positions, if necessary, the gaming system determines the winning symbol combinations for the play of the game based on those selected active symbol display positions. For example, the gaming system can select, configure, or reconfigure the symbols display that are in the form of reels and then displays those reels for the play of the game in association with the active symbol display positions. Each of the displayed plurality of reels include a plurality of different symbols on such reel.

In various such embodiments, responsive to an occurrence of the game play triggering event, for each symbol display and specifically for each reel associated with an active symbol display position, the gaming system: (i) causes that reel to spin; (ii) randomly determines one or more symbols on that reel to display at the symbol display position(s) associated with that reel; and (iii) causes that reel to stop to display the randomly determined symbol(s) at the active symbol display position(s) associated with that reel. The gaming system then performs the symbol evaluation for the play of the game taking into account the active symbol display positions and the possible winning symbol combinations. The gaming system displays indications of any evaluations based on the active symbol display positions for and the respective symbol evaluations for the play of the game, and the gaming system determines and displays any awards for the play of the game.

While various embodiments described below are directed to the gaming system providing the active symbol display position feature displayed by an electronic gaming machine ("EGM") in the form of a slot machine, it should be appreciated that such embodiments can additionally or alternatively be employed in association with a video poker machine, a video lottery terminal, a terminal associated with an electronic table game, a terminal associated with a live table game, a video keno machine, a video bingo machine, a sports betting terminal that also offers sports betting opportunities, a personal gaming device such as a desktop computer, a laptop computer, a tablet computer or computing device, a personal digital assistant, a mobile telephone, and/or other mobile computing device that offer plays of games (and in certain instances, sports betting opportunities).

While various embodiments described herein relate to the gaming system displaying a credit balance, a wager, and any awards as an amount of monetary credits or currency, in other embodiments, one or more of such credit balance, such wager, and any awards provided to such a player can be for non-monetary credits, promotional credits, and/or player tracking points or credits.

FIG. 1 is a flowchart of an example process or method 100 of operating the gaming system of the present disclosure. In various embodiments, the process is represented by a set of instructions stored in one or more memories and executed by one or more processors. Although the process is described with reference to the flowchart shown in FIG. 1, many other

processes of performing the acts associated with this illustrated process may be employed. For example, the order of certain of the illustrated blocks or diamonds can be changed, certain of the illustrated blocks or diamonds can be optional, or certain of the illustrated blocks or diamonds may not be employed.

Responsive to an occurrence of a slot game triggering event, the gaming system triggers a play of a slot game, as indicated in block 102. In various embodiments wherein the slot game is provided as a primary wagering slot game, the slot game triggering event occurs upon a placement of a wager by a player on the play of the slot game. In various embodiments wherein the slot game is a secondary game, the slot game triggering event occurs based on a displayed event associated with a play of an underlying primary game. In various embodiments wherein the slot game is a secondary game, the slot game triggering event occurs based on an event independent of any displayed event associated with the play of the underlying primary game.

The gaming system displays the symbol display matrix and displays an indication of the different symbol display position selection options for the play of the slot game, as indicated in block 104. In various embodiments, the symbol display position selection options can be static or can vary for two or more or each play of the slot game. In various embodiments, each of the different symbol display position selection options relates to different groups of symbol display positions that can be active and non-active and that the player can select for that play of the slot game. The gaming system is configured to receive a player selection of one of the options for the play of the slot game. In one example embodiment, the different symbol display position selection options include: (a) three active columns of symbol display positions; (b) four active columns of symbol display positions; and (c) five active columns of symbol display positions.

The gaming system displays indications of the player active and non-active symbol display positions (if any) for the play of the slot game, as indicated in block 106. The display can be directly associated with each of the displayed reels, the symbol display positions, or in another suitable manner.

The gaming system randomly determines and then displays a plurality of symbols on the plurality of reels at the plurality of active symbol display positions associated with the reels for the play of the slot game, as indicated in block 108.

The gaming system determines any awards associated with the symbols on the reels displayed at the active symbol display positions for the play of the game, as indicated in block 110. In various embodiments, the gaming system determines any awards associated with the displayed symbols on the reels based on: (1) the payable for the play of the slot game; (2) the wager for the play of the slot game; (3) the selected active symbol display positions for the play of the slot game; and (4) one or more different symbol evaluation methods for the play of the slot game. The symbols evaluation methods can, for example, include a payable symbol evaluation method or a ways to win symbol evaluation method.

FIGS. 2A, 2B, and 2C illustrate example screen displays of portions of a first example play of a primary wagering slot game of one example embodiment of the present disclosure on a gaming system such as an EGM (such as an EGM described above or below). In this example, the EGM displays via display device 200 (such as a display device described below) adjacently arranged video reels 210, 220,

230, 240, and 250 and different symbols (not labeled) on the video reels **210, 220, 230, 240, and 250**. For each play of the slot game, the reels **210, 220, 230, 240, and 250** are configured to spin, stop, and after stopping display a plurality of the different symbols on the reels at respective sets of symbol display positions **215a to 215d, 225a to 225d, 2315a to 235d, 245a to 245d, and 255a to 255d** respectively associated with the reels **210, 220, 230, 240, and 250**. It should be appreciated that these displays can vary in accordance with the present disclosure, and that the EGM can cause the display of additional game play related information. For example, the EGM can cause the display device **200** to display: (1) a credit meter that displays the player's credit balance; (2) a wager display that displays any wagers placed on any plays of the primary wagering slot game; (3) a win display that displays any awards won for each play of the primary wagering slot game; and (4) a message box configured to display messages before, during, or after each play of each slot game. In various embodiments, the EGM indicates the player's credit balance, the player's wager, and any awards provided to the player in the form of amounts of credits; however, it should be appreciated that such indications can alternatively or additionally be made in the form of amounts of currency, points, or the like.

FIG. 2A shows an example screen display for a play of primary wagering slot game that has been triggered on the EGM. The EGM displays a plurality of different active symbol display position options in the form player selectable input buttons **260, 262, 264, 266, and 268**, that the player can individually select to make each group of symbol display positions active. In this example, each group is a column of symbol display positions. The EGM also displays an indication that the player can select any of the columns of symbol display positions to be active. In this example, the player's selection of the respective columns, also selected the respective volatility for the play of the game. In this example, the player selects four out of the five columns of symbol display positions to be active for this play of the game.

FIG. 2B shows an example screen display at a subsequent point in the play of the slot game where the player has selected the first four columns of symbol display positions to be active for this play of the game, the EGM has displayed an indication that the first four columns of symbol display positions are active and displayed an indication that the last or fifth column of symbol display positions are not active for this play of the slot game. It should be appreciated that the player could have alternatively selected the columns such as by: (1) selecting the first column of symbol display positions to be non-active and the last four columns of symbol display positions to be active; (2) selecting the third column of symbol display positions to be non-active and the first two and last two columns of symbol display positions to be active; (3) selecting the second column of symbol display positions to be non-active and the first, third, fourth, and fifth columns of symbol display positions to be active; or (4) selecting the fourth column of symbol display positions to be non-active and the first, second, third, and fifth columns of symbol display positions to be active.

FIG. 2C shows an example screen display at a subsequent point in the play of the game where the EGM has randomly determined and displayed a plurality of symbols on the video reels **210, 220, 230, and 240** (but not on reel **250**) at the respective symbol display positions. FIG. 2C shows the EGM displaying an indication that the EGM has evaluated the displayed symbols for winning combinations along the paylines, wherein the evaluation includes the symbols on the

selected symbol display positions. For this evaluation, since the player selected four active columns of symbol display positions, the EGM evaluates each of the paylines PL1, PL2, PL3, and PL4 for any combination of four matching symbols along such payline which results in a winning symbol combination. For this evaluation, there is no winning symbol combination along payline PL1, there is a winning symbol combination along payline PL2 of 4 M3 symbols that results in an award of 2000 credits, there are no winning symbol combinations along payline PL3, and there is a winning symbol combinations along payline PL4 of four A symbols that results in an award of 800 credits. It should be appreciated that in this example that there are no winning symbol combinations along payline PL3 because there are only 3 M2 symbols along that payline instead of the required 4 symbols for the winning symbol combination based on the selected active symbol display positions. Thus, the selection of the quantity of active symbol display positions also causes a selection of what symbol combinations are winning symbol combinations and thus of the volatility of the game.

FIGS. 3A, 3B, and 3C illustrate example screen displays of portions of a second example play of a primary wagering slot game of one example embodiment of the present disclosure on a gaming system such as an EGM (such as an EGM described above or below). In this example, the EGM displays via display device **200** (such as a display device described below) adjacently arranged video reels **210, 220, 230, 240, and 250** and different symbols (not labeled) on the video reels **210, 220, 230, 240, and 250**. For each play of the slot game, the reels **210, 220, 230, 240, and 250** are configured to spin, stop, and after stopping display a plurality of the different symbols on the reels at respective sets of symbol display positions **215a to 215d, 225a to 225d, 2315a to 235d, 245a to 245d, and 255a to 255d** respectively associated with the reels **210, 220, 230, 240, and 250**. It should be appreciated that these displays can vary in accordance with the present disclosure, and that the EGM can cause the display of additional game play related information. For example, the EGM can cause the display device **200** to display: (1) a credit meter that displays the player's credit balance; (2) a wager display that displays any wagers placed on any plays of the primary wagering slot game; (3) a win display that displays any awards won for each play of the primary wagering slot game; and (4) a message box configured to display messages before, during, or after each play of each slot game. In various embodiments, the EGM indicates the player's credit balance, the player's wager, and any awards provided to the player in the form of amounts of credits; however, it should be appreciated that such indications can alternatively or additionally be made in the form of amounts of currency, points, or the like.

FIG. 3A shows an example screen display for a play of primary wagering slot game that has been triggered on the EGM. The EGM displays a plurality of different active symbol display position options in the form player selectable input buttons **260, 262, 264, 266, and 268**, that the player can individually select to make each group of symbol display positions active. In this example, each group is a column of symbol display positions. The EGM also displays an indication that the player can select any of the columns of symbol display positions to be active. In this example, the player selects three out of the five columns of symbol display positions to be active for this play of the game.

FIG. 3B shows an example screen display at a subsequent point in the play of the slot game where the player has selected the first three columns of symbol display positions to be active for this play of the game, the EGM has displayed

an indication that the first three columns of symbol display positions are active and displayed an indication that the last two columns of symbol display positions are not active for this play of the slot game. It should be appreciated that the player could have alternatively selected the columns such as by: (1) selecting the first two columns of symbol display positions to be non-active and the last three columns of symbol display positions to be active; (2) selecting the first and last columns of symbol display positions to be non-active and the second third, and fourth middle three columns of symbol display positions to be active; or (3) selecting the second and fourth columns of symbol display positions to be non-active and the first, third, and fifth columns of symbol display positions to be active.

FIG. 3C shows an example screen display at a subsequent point in the play of the game where the EGM has randomly determined and displayed a plurality of symbols on the video reels **210**, **220**, and **230** at the respective symbol display positions (but not on reels **240** or **250** because those reels are not associated with active symbol display positions). FIG. 3C shows the EGM displaying an indication that the EGM has evaluated the displayed symbols for winning combinations along the paylines, wherein the evaluation includes the symbols on the selected symbol display positions. For this evaluation, since the player selected three active columns of symbol display positions, the EGM evaluates each of the paylines PL1, PL2, PL3, and PL4 for any combination of three matching symbols along such payline which results in a winning symbol combination. For this evaluation, there is a winning symbol combination along payline PL1 of 3 7 symbols that results in an award of 200 credits, there is a winning symbol combination along payline PL2 of 3 M3 symbols that results in an award of 500 credits, there are no winning symbol combinations along payline PL3, and there is a winning symbol combination along payline PL4 of 3 K symbols that results in an award of 300 credits. It should be appreciated that in this example that there are no winning symbol combinations along payline PL3 because there are only 2 M2 symbols along that payline instead of the required 3 symbols for the winning symbol combination based on the selected active symbol display positions. Thus, the selection of the quantity of active symbol display positions also causes a selection of what symbol combinations are winning symbol combinations and thus of the volatility of the game.

FIGS. 4A, 4B, and 4C illustrate example screen displays of portions of an example play of a different primary wagering slot game of one example embodiment of the present disclosure on a gaming system such as an EGM (such as an EGM described above or below). In this example, the EGM displays via display device **200** (such as a display device described below) adjacently arranged video reels **210**, **220**, **230**, **240**, and **250** and different symbols (not labeled) on the video reels **210**, **220**, **230**, **240**, and **250**. For each play of the slot game, the reels **210**, **220**, **230**, **240**, and **250** are configured to spin, stop, and after stopping display a plurality of the different symbols on the reels at respective sets of symbol display positions **215a** to **215d**, **225a** to **225d**, **2315a** to **235d**, **245a** to **245d**, and **255a** to **255d** respectively associated with the reels **210**, **220**, **230**, **240**, and **250**. It should be appreciated that these displays can vary in accordance with the present disclosure, and that the EGM can cause the display of additional game play related information. For example, the EGM can cause the display device **200** to display: (1) a credit meter that displays the player's credit balance; (2) a wager display that displays any wagers placed on any plays of the primary wagering slot game; (3)

a win display that displays any awards won for each play of the primary wagering slot game; and (4) a message box configured to display messages before, during, or after each play of each slot game. In various embodiments, the EGM indicates the player's credit balance, the player's wager, and any awards provided to the player in the form of amounts of credits; however, it should be appreciated that such indications can alternatively or additionally be made in the form of amounts of currency, points, or the like.

FIG. 4A shows an example screen display for a play of primary wagering slot game that has been triggered on the EGM. The EGM displays a plurality of different active symbol display position options in the form player selectable input buttons **270**, **272**, **274**, and **276** that the player can individually select to make each group of symbol display positions active. In this example, each group is a row of symbol display positions. The EGM also displays an indication that the player can select any of the row of symbol display positions to be active. In this example, the player selects three out of the four rows of symbol display positions to be active for this play of the game.

FIG. 4B shows an example screen display at a subsequent point in the play of the slot game where the player has selected rows A, B, and D of symbol display positions to be active for this play of the game, the EGM has displayed an indication that the rows A, B, and D of symbol display positions are active, and displayed an indication that row C of symbol display positions is not active for this play of the slot game. It should be appreciated that the player could have alternatively selected the row such as by: (1) selecting the first row of symbol display positions to be non-active and the last three rows of symbol display positions to be active; (2) selecting the second row of symbol display positions to be non-active and the first and last two rows of symbol display positions to be active; or (3) selecting the last row of symbol display positions to be non-active and the first three rows of symbol display positions to be active.

FIG. 4C shows an example screen display at a subsequent point in the play of the game where the EGM has randomly determined and displayed a plurality of symbols on the video reels **210**, **220**, **230**, **240**, and **250** at the respective symbol display positions except for at the symbol display positions associated with the non-active row C. FIG. 4C shows the EGM displaying an indication that the EGM has evaluated the displayed symbols for winning combinations along the paylines, wherein the evaluation includes the symbols on the selected symbol display positions. For this evaluation, since the player selected three rows of active symbol display positions, the EGM evaluates each of the paylines PL1, PL2, and PL4 for any combination of five matching symbols along such payline which results in a winning symbol combination. For this evaluation, there is no winning symbol combination along payline PL1, there is a winning symbol combination along payline PL2 of 5 M3 symbols that results in an award of 3000 credits, there are no winning symbol combinations along payline PL3, and there is a winning symbol combinations along payline PL4 of five A symbols that results in an award of 1000 credits. It should be appreciated that in this example that there are no winning symbol combinations along payline PL3 because the player made the symbol display positions along that payline non-active. Thus, the selection of the quantity of active symbol display positions also causes a selection of what symbol combinations are winning symbol combinations and the respective active paylines, and thus of the volatility of the game.

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It should be appreciated from the above examples that in various embodiments, the quantity of active symbol positions (such as along a payline) partially determine what symbol combinations are winning symbol combinations and thus the volatility for the play of the game. For example, in certain embodiments, if five active symbol positions along a payline are selected, then a winning symbol combination along that payline includes five matching symbols along that payline (instead of one, two, three, or four matching symbols along that payline). In such embodiments, three or four matching symbols which would in conventional slot game paytables be winning symbol combinations, would not be winning symbol combinations. Thus, the volatility of the game is selected based on the selection of the active symbol display positions and the corresponding selection of the winning symbol combinations.

It should be appreciated from the above that various embodiments of the present disclosure provide a gaming system and methods that: (1) cause a display of symbol displays including different symbols; (2) cause a display of an indication of selectable symbol display positions associated with the symbol displays; (3) cause a display of an indication of a selected plurality of the symbol display positions as being active symbol display positions for the play of the game; (4) cause a display of a plurality of the different symbols on the symbol displays displayed at the plurality of active symbol display positions; and (5) cause a display of an indication of a first quantity of same symbols at a plurality of the active symbol display positions, wherein whether the first quantity of same symbols is a winning symbol combination depends on the symbol display positions selected as the active symbol display positions, wherein if a first quantity of symbol display positions are active symbol display positions, the first quantity of same symbols at the active symbol display positions forms the winning symbol combination, and wherein if a greater second quantity of symbol display positions are active symbol display positions, the first quantity of same symbols at the active symbol display positions does not form the winning symbol combination. Various such embodiments further cause the active symbol display positions to be less than all of the symbol display positions. Various such embodiments further cause a display of an indication of different probabilities of winning based on how many symbol display positions are selected as active symbol display positions.

Various such embodiments further cause the display of the indication of selectable symbol display positions to include an indication of different selectable columns of the symbol display positions as active columns of symbol display positions. Various such embodiments further cause a display of an indication of different probabilities of winning based on how many columns of symbol display positions are selected as active columns of symbol display positions.

Various such embodiments further cause the display of the indication of selectable symbol display positions to include an indication of different selectable rows of the symbol display positions as active rows of symbol display positions. Various such embodiments further cause a display of an indication of different probabilities of winning based on how many rows of symbol display positions are selected as active rows of symbol display positions.

It should further be appreciated from the above that various embodiments of the present disclosure provide a gaming system and methods that: (1) cause a display of symbol displays including different symbols; (2) cause a display of an indication of selectable symbol display posi-

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tions associated with the symbol display; (3) for a first play of a game: (i) cause a display of an indication of all of the symbol display positions as being first active symbol display positions for the first play of the game; (ii) cause a display of a first plurality of the different symbols on the symbol displays displayed at the first active symbol display positions; and (iii) cause a display of an indication of a first quantity of same symbols at the first active symbol display positions, wherein whether the first quantity of same symbols is a winning symbol combination depends on the symbol display positions selected as the first active symbol display positions, wherein the first quantity of same symbols at the first active symbol display positions does not form the winning symbol combination; and (4) for a second play of the game: (i) cause a display of an indication of less than all of the symbol display positions as being second active symbol display positions for the second play of the game; (ii) cause a display of a second plurality of the different symbols on the symbol displays displayed at the second active symbol display positions; and (iii) cause a display of an indication of the first quantity of same symbols at the second active symbol display positions, wherein whether the first quantity of same symbols is the winning symbol combination depends on the symbol display positions selected as the second active symbol display positions, wherein the first quantity of same symbols at the second active symbol display positions form the winning symbol combination. Various such embodiments further cause a display of an indication of different probabilities of winning based on how many symbol display positions are selected as active symbol display positions.

Various such embodiments further cause the display of the indication of selectable symbol display positions to include an indication of different selectable columns of the symbol display positions as active columns of symbol display positions. Various such embodiments further cause a display of an indication of different probabilities of winning based on how many columns of symbol display positions are selected as active columns of symbol display positions.

Various such embodiments further cause the display of the indication of selectable symbol display positions to include an indication of different selectable rows of the symbol display positions as active rows of symbol display positions. Various such embodiments further cause a display of an indication of different probabilities of winning based on how many rows of symbol display positions are selected as active rows of symbol display positions.

It should further be appreciated from the above that various embodiments of the present disclosure provide a gaming system and methods that: (1) cause a display of symbol displays including different symbols; (2) cause a display of an indication of selectable symbol display positions associated with the symbol display; (3) for a first play of a game: (i) cause a display of an indication of a first quantity of the symbol display positions as being first active symbol display positions for the first play of the game; (ii) cause a display of a first plurality of the different symbols on the symbol displays displayed at the first active symbol display positions; and (iii) cause a display of an indication of a first quantity of same symbols at the first active symbol display positions, wherein whether the first quantity of same symbols is a winning symbol combination depends on the symbol display positions selected as the first active symbol display positions, wherein the first quantity of same symbols at the first active symbol display positions does not form the winning symbol combination; and (4) for a second play of the game: (i) cause a display of an indication of a second

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quantity of the symbol display positions as being second active symbol display positions for the second play of the game; (ii) cause a display of a second plurality of the different symbols on the symbol displays displayed at the second active symbol display positions; and (iii) cause a display of an indication the first quantity of same symbols at the second active symbol display positions, wherein whether the first quantity of same symbols is the winning symbol combination depends on the symbol display positions selected as the second active symbol display positions, wherein the first quantity of same symbols at the second active symbol display positions form the winning symbol combination. Various such embodiments further cause a display of an indication of different probabilities of winning based on how many symbol display positions are selected as active symbol display positions. Various such embodiments further cause the display of the indication of selectable symbol display positions to include an indication of different selectable columns of the symbol display positions as active columns of symbol display positions. Various such embodiments further cause a display of an indication of different probabilities of winning based on how many columns of symbol display positions are selected as active columns of symbol display positions.

Various such embodiments further cause the display of the indication of selectable symbol display positions to include an indication of different selectable rows of the symbol display positions as active rows of symbol display positions. Various such embodiments further cause a display of an indication of different probabilities of winning based on how many rows of symbol display positions are selected as active rows of symbol display positions.

Various such embodiments further cause the display of the indication of selectable symbol display positions to include an indication of different selectable rows of the symbol display positions as active rows of symbol display positions. Various such embodiments further cause a display of an indication of different probabilities of winning based on how many rows of symbol display positions are selected as active rows of symbol display positions.

It should further be appreciated that in different embodiments, one or more of various different determinations disclosed herein is/are predetermined, randomly determined, randomly determined based on one or more weighted percentages, determined based on a generated symbol or symbol combination, determined independent of a generated symbol or symbol combination, determined based on a random determination by the central controller, determined independent of a random determination by the central controller, determined based on a random determination at the gaming system, determined independent of a random determination at the gaming system, determined based on at least one play of at least one game, determined independent of at least one play of at least one game, determined based on a user's selection, determined independent of a user's selection, determined based on one or more side wagers placed, determined independent of one or more side wagers placed, determined based on the user's primary game wager, determined independent of the user's primary game wager, determined based on time (such as the time of day), determined independent of time (such as the time of day), determined based on an amount of coin-in accumulated in one or more pools, determined independent of an amount of coin-in accumulated in one or more pools, determined based on a status of the user (i.e., a player tracking status), determined independent of a status of the user (i.e., a player tracking status), determined based on one or more other determinations disclosed herein, determined independent of

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any other determination disclosed herein or determined based on any other suitable method or criteria.

Gaming Systems

The above-described embodiments of the present disclosure may be implemented in accordance with or in conjunction with one or more of a variety of different types of gaming systems, such as, but not limited to, those described below.

The present disclosure contemplates a variety of different gaming systems each having one or more of a plurality of different features, attributes, or characteristics. A "gaming system" as used herein refers to various configurations of: (a) one or more servers; (b) one or more electronic gaming machines such as those located on a casino floor; and/or (c) one or more personal gaming devices. Thus, in various embodiments, the gaming system of the present disclosure includes: (a) one or more electronic gaming machines in combination with one or more servers; (b) one or more personal gaming devices in combination with one or more servers; (c) one or more personal gaming devices in combination with one or more electronic gaming machines; (d) one or more personal gaming devices, one or more electronic gaming machines, and one or more servers in combination with one another; (e) a single electronic gaming machine; (f) a plurality of electronic gaming machines in combination with one another; (g) a single personal gaming device; (h) a plurality of personal gaming devices in combination with one another; (i) a single server; and/or (j) a plurality of servers in combination with one another. For brevity and clarity and unless specifically stated otherwise, "EGM" as used herein represents one EGM or a plurality of EGMs, "personal gaming device" as used herein represents one personal gaming device or a plurality of personal gaming devices, and "server" as used herein represents one server or a plurality of servers.

As noted above, in various embodiments, the gaming system includes an EGM (or personal gaming device) in combination with a server. In such embodiments, the EGM (or personal gaming device) is configured to communicate with the server through a data network or remote communication link. In certain such embodiments, the EGM (or personal gaming device) is configured to communicate with another EGM (or personal gaming device) through the same data network or remote communication link or through a different data network or remote communication link. For example, the gaming system includes a plurality of EGMs that are each configured to communicate with a server through a data network.

In certain embodiments in which the gaming system includes an EGM (or personal gaming device) in combination with a server, the server is any suitable computing device (such as a server) that includes at least one processor and at least one memory device or data storage device. As further described herein, the EGM (or personal gaming device) includes at least one EGM (or personal gaming device) processor configured to transmit and receive data or signals representing events, messages, commands, or any other suitable information between the EGM (or personal gaming device) and the server. The at least one processor of that EGM (or personal gaming device) is configured to execute the events, messages, or commands represented by such data or signals in conjunction with the operation of the EGM (or personal gaming device). Moreover, the at least one processor of the server is configured to transmit and receive data or signals representing events, messages, com-

mands, or any other suitable information between the server and the EGM (or personal gaming device). The at least one processor of the server is configured to execute the events, messages, or commands represented by such data or signals in conjunction with the operation of the server. One, more than one, or each of the functions of the server may be performed by the at least one processor of the EGM (or personal gaming device). Further, one, more than one, or each of the functions of the at least one processor of the EGM (or personal gaming device) may be performed by the at least one processor of the server.

In certain such embodiments, computerized instructions for controlling any games (such as any primary or base games and/or any secondary or bonus games) displayed by the EGM (or personal gaming device) are executed by the server. In such "thin client" embodiments, the server remotely controls any games (or other suitable interfaces) displayed by the EGM (or personal gaming device), and the EGM (or personal gaming device) is utilized to display such games (or suitable interfaces) and to receive one or more inputs or commands. In other such embodiments, computerized instructions for controlling any games displayed by the EGM (or personal gaming device) are communicated from the server to the EGM (or personal gaming device) and are stored in at least one memory device of the EGM (or personal gaming device). In such "thick client" embodiments, the at least one processor of the EGM (or personal gaming device) executes the computerized instructions to control any games (or other suitable interfaces) displayed by the EGM (or personal gaming device).

In various embodiments in which the gaming system includes a plurality of EGMs (or personal gaming devices), one or more of the EGMs (or personal gaming devices) are thin client EGMs (or personal gaming devices) and one or more of the EGMs (or personal gaming devices) are thick client EGMs (or personal gaming devices). In other embodiments in which the gaming system includes one or more EGMs (or personal gaming devices), certain functions of one or more of the EGMs (or personal gaming devices) are implemented in a thin client environment, and certain other functions of one or more of the EGMs (or personal gaming devices) are implemented in a thick client environment. In one such embodiment in which the gaming system includes an EGM (or personal gaming device) and a server, computerized instructions for controlling any primary or base games displayed by the EGM (or personal gaming device) are communicated from the server to the EGM (or personal gaming device) in a thick client configuration, and computerized instructions for controlling any secondary or bonus games or other functions displayed by the EGM (or personal gaming device) are executed by the server in a thin client configuration.

In certain embodiments in which the gaming system includes: (a) an EGM (or personal gaming device) configured to communicate with a server through a data network; and/or (b) a plurality of EGMs (or personal gaming devices) configured to communicate with one another through a data network, the data network is a local area network (LAN) in which the EGMs (or personal gaming devices) are located substantially proximate to one another and/or the server. In one example, the EGMs (or personal gaming devices) and the server are located in a gaming establishment or a portion of a gaming establishment.

In other embodiments in which the gaming system includes: (a) an EGM (or personal gaming device) configured to communicate with a server through a data network; and/or (b) a plurality of EGMs (or personal gaming devices)

configured to communicate with one another through a data network, the data network is a wide area network (WAN) in which one or more of the EGMs (or personal gaming devices) are not necessarily located substantially proximate to another one of the EGMs (or personal gaming devices) and/or the server. For example, one or more of the EGMs (or personal gaming devices) are located: (a) in an area of a gaming establishment different from an area of the gaming establishment in which the server is located; or (b) in a gaming establishment different from the gaming establishment in which the server is located. In another example, the server is not located within a gaming establishment in which the EGMs (or personal gaming devices) are located. In certain embodiments in which the data network is a WAN, the gaming system includes a server and an EGM (or personal gaming device) each located in a different gaming establishment in a same geographic area, such as a same city or a same state. Gaming systems in which the data network is a WAN are substantially identical to gaming systems in which the data network is a LAN, though the quantity of EGMs (or personal gaming devices) in such gaming systems may vary relative to one another.

In further embodiments in which the gaming system includes: (a) an EGM (or personal gaming device) configured to communicate with a server through a data network; and/or (b) a plurality of EGMs (or personal gaming devices) configured to communicate with one another through a data network, the data network is an internet (such as the Internet) or an intranet. In certain such embodiments, an Internet browser of the EGM (or personal gaming device) is usable to access an Internet game page from any location where an Internet connection is available. In one such embodiment, after the EGM (or personal gaming device) accesses the Internet game page, the server identifies a user before enabling that user to place any wagers on any plays of any wagering games. In one example, the server identifies the user by requiring a user account of the user to be logged into via an input of a unique username and password combination assigned to the user. The server may, however, identify the user in any other suitable manner, such as by validating a player tracking identification number associated with the user; by reading a player tracking card or other smart card inserted into a card reader (as described below); by validating a unique user identification number associated with the user by the server; or by identifying the EGM (or personal gaming device), such as by identifying the MAC address or the IP address of the Internet facilitator. In various embodiments, once the server identifies the user, the server enables placement of one or more wagers on one or more plays of one or more primary or base games and/or one or more secondary or bonus games and displays those plays via the Internet browser of the EGM (or personal gaming device).

The server and the EGM (or personal gaming device) are configured to connect to the data network or remote communications link in any suitable manner. In various embodiments, such a connection is accomplished via a conventional phone line or other data transmission line, a digital subscriber line (DSL), a T-1 line, a coaxial cable, a fiber optic cable, a wireless or wired routing device, a mobile communications network connection (such as a cellular network or mobile Internet network), or any other suitable medium. The expansion in the quantity of computing devices and the quantity and speed of Internet connections in recent years increases opportunities for users to use a variety of EGMs (or personal gaming devices) to play games from an ever-increasing quantity of remote sites. Additionally, the enhanced bandwidth of digital wireless communications

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may render such technology suitable for some or all communications, particularly if such communications are encrypted. Higher data transmission speeds may be useful for enhancing the sophistication and response of the display and interaction with users.

FIG. 5 is a block diagram of an example EGM **1000** and FIGS. 6A and 6B include two different example EGMs **2000a** and **2000b**. The EGMs **1000**, **2000a**, and **2000b** are merely example EGMs, and different EGMs may be implemented using different combinations of the components shown in the EGMs **1000**, **2000a**, and **2000b**. Although the below refers to EGMs, in various embodiments personal gaming devices (such as personal gaming device **2000c** of FIG. 6C) may include some or all of the below components.

In these embodiments, the EGM **1000** includes a master gaming controller **1012** configured to communicate with and to operate with a plurality of peripheral devices **1022**.

The master gaming controller **1012** includes at least one processor **1010**. The at least one processor **1010** is any suitable processing device or set of processing devices, such as a microprocessor, a microcontroller-based platform, a suitable integrated circuit, or one or more application-specific integrated circuits (ASICs), configured to execute software enabling various configuration and reconfiguration tasks, such as: (1) communicating with a remote source (such as a server that stores authentication information or game information) via a communication interface **1006** of the master gaming controller **1012**; (2) converting signals read by an interface to a format corresponding to that used by software or memory of the EGM; (3) accessing memory to configure or reconfigure game parameters in the memory according to indicia read from the EGM; (4) communicating with interfaces and the peripheral devices **1022** (such as input/output devices); and/or (5) controlling the peripheral devices **1022**. In certain embodiments, one or more components of the master gaming controller **1012** (such as the at least one processor **1010**) reside within a housing of the EGM (described below), while in other embodiments at least one component of the master gaming controller **1012** resides outside of the housing of the EGM.

The master gaming controller **1012** also includes at least one memory device **1016**, which includes: (1) volatile memory (e.g., RAM **1009**, which can include non-volatile RAM, magnetic RAM, ferroelectric RAM, and any other suitable forms); (2) non-volatile memory **1019** (e.g., disk memory, FLASH memory, EPROMs, EEPROMs, memristor-based non-volatile solid-state memory, etc.); (3) unalterable memory (e.g., EPROMs **1008**); (4) read-only memory; and/or (5) a secondary memory storage device **1015**, such as a non-volatile memory device, configured to store gaming software related information (the gaming software related information and the memory may be used to store various audio files and games not currently being used and invoked in a configuration or reconfiguration). Any other suitable magnetic, optical, and/or semiconductor memory may operate in conjunction with the EGM of the present disclosure. In certain embodiments, the at least one memory device **1016** resides within the housing of the EGM (described below), while in other embodiments at least one component of the at least one memory device **1016** resides outside of the housing of the EGM. In these embodiments, any combination of one or more computer readable media may be utilized. The computer readable media may be a computer readable signal medium or a computer readable storage medium. A computer readable storage medium may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, or semiconductor system, apparatus, or

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device, or any suitable combination of the foregoing. More specific examples (a non-exhaustive list) of the computer readable storage medium would include the following: a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), an appropriate optical fiber with a repeater, a portable compact disc read-only memory (CD-ROM), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any tangible medium that can contain or store a program for use by or in connection with an instruction execution system, apparatus, or device.

A computer readable signal medium may include a propagated data signal with computer readable program code embodied therein, for example, in baseband or as part of a carrier wave. Such a propagated signal may take any of a variety of forms, including, but not limited to, electromagnetic, optical, or any suitable combination thereof. A computer readable signal medium may be any computer readable medium that is not a computer readable storage medium and that can communicate, propagate, or transport a program for use by or in connection with an instruction execution system, apparatus, or device. Program code embodied on a computer readable signal medium may be transmitted using any appropriate medium, including but not limited to wireless, wireline, optical fiber cable, RF, etc., or any suitable combination of the foregoing.

The at least one memory device **1016** is configured to store, for example: (1) configuration software **1014**, such as all the parameters and settings for a game playable on the EGM; (2) associations **1018** between configuration indicia read from an EGM with one or more parameters and settings; (3) communication protocols configured to enable the at least one processor **1010** to communicate with the peripheral devices **1022**; and/or (4) communication transport protocols (such as TCP/IP, USB, Firewire, IEEE1394, Bluetooth, IEEE 802.11x (IEEE 802.11 standards), hipernlan/2, HomeRF, etc.) configured to enable the EGM to communicate with local and non-local devices using such protocols. In one implementation, the master gaming controller **1012** communicates with other devices using a serial communication protocol. A few non-limiting examples of serial communication protocols that other devices, such as peripherals (e.g., a bill validator or a ticket printer), may use to communicate with the master game controller **1012** include USB, RS-232, and Netplex (a proprietary protocol developed by IGT).

As will be appreciated by one skilled in the art, aspects of the present disclosure may be illustrated and described herein in any of a number of patentable classes or context including any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof. Accordingly, aspects of the present disclosure may be implemented entirely hardware, entirely software (including firmware, resident software, microcode, etc.) or combining software and hardware implementation that may all generally be referred to herein as a "circuit," "module," "component," or "system." Furthermore, aspects of the present disclosure may take the form of a computer program product embodied in one or more computer readable media having computer readable program code embodied thereon.

Computer program code for carrying out operations for aspects of the present disclosure may be written in any combination of one or more programming languages,

including an object oriented programming language such as Java, Scala, Smalltalk, Eiffel, JADE, Emerald, C++, C#, VB.NET, Python or the like, conventional procedural programming languages, such as the "C" programming language, Visual Basic, Fortran 2003, Perl, COBOL 2002, PHP, ABAP, dynamic programming languages such as Python, Ruby and Groovy, or other programming languages. The program code may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider) or in a cloud computing environment or offered as a service such as a Software as a Service (SaaS).

Aspects of the present disclosure are described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatuses (systems) and computer program products according to embodiments of the disclosure. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable instruction execution apparatus, create a mechanism for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

These computer program instructions may also be stored in a computer readable medium that when executed can direct a computer, other programmable data processing apparatus, or other devices to function in a particular manner, such that the instructions when stored in the computer readable medium produce an article of manufacture including instructions which when executed, cause a computer to implement the function/act specified in the flowchart and/or block diagram block or blocks. The computer program instructions may also be loaded onto a computer, other programmable instruction execution apparatus, or other devices to cause a series of operational steps to be performed on the computer, other programmable apparatuses or other devices to produce a computer implemented process such that the instructions which execute on the computer or other programmable apparatus provide processes for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

In certain embodiments, the at least one memory device **1016** is configured to store program code and instructions executable by the at least one processor of the EGM to control the EGM. The at least one memory device **1016** of the EGM also stores other operating data, such as image data, event data, input data, random number generators (RNGs) or pseudo-RNGs, paytable data or information, and/or applicable game rules that relate to the play of one or more games on the EGM. In various embodiments, part, or all of the program code and/or the operating data described above is stored in at least one detachable or removable memory device including, but not limited to, a cartridge, a disk, a CD ROM, a DVD, a USB memory device, or any other suitable non-transitory computer readable medium. In

certain such embodiments, an operator (such as a gaming establishment operator) and/or a user uses such a removable memory device in an EGM to implement at least part of the present disclosure. In other embodiments, part, or all of the program code and/or the operating data is downloaded to the at least one memory device of the EGM through any suitable data network described above (such as an Internet or intranet).

The at least one memory device **1016** also stores a plurality of device drivers **1042**. Examples of different types of device drivers include device drivers for EGM components and device drivers for the peripheral components **1022**. Typically, the device drivers **1042** utilize various communication protocols that enable communication with a particular physical device. The device driver abstracts the hardware implementation of that device. For example, a device driver may be written for each type of card reader that could potentially be connected to the EGM. Non-limiting examples of communication protocols used to implement the device drivers include Netplex, USB, Serial, Ethernet 175, Firewire, I/O debouncer, direct memory map, serial, PCI, parallel, RF, Bluetooth™, near-field communications (e.g., using near-field magnetics), 802.11 (WiFi), etc. In one embodiment, when one type of a particular device is exchanged for another type of the particular device, the at least one processor of the EGM loads the new device driver from the at least one memory device to enable communication with the new device. For instance, one type of card reader in the EGM can be replaced with a second different type of card reader when device drivers for both card readers are stored in the at least one memory device.

In certain embodiments, the software units stored in the at least one memory device **1016** can be upgraded as needed. For instance, when the at least one memory device **1016** is a hard drive, new games, new game options, new parameters, new settings for existing parameters, new settings for new parameters, new device drivers, and new communication protocols can be uploaded to the at least one memory device **1016** from the master game controller **1012** or from some other external device. As another example, when the at least one memory device **1016** includes a CD/DVD drive including a CD/DVD configured to store game options, parameters, and settings, the software stored in the at least one memory device **1016** can be upgraded by replacing a first CD/DVD with a second CD/DVD. In yet another example, when the at least one memory device **1016** uses flash memory **1019** or EPROM **1008** units configured to store games, game options, parameters, and settings, the software stored in the flash and/or EPROM memory units can be upgraded by replacing one or more memory units with new memory units that include the upgraded software. In another embodiment, one or more of the memory devices, such as the hard drive, may be employed in a game software download process from a remote software server.

In some embodiments, the at least one memory device **1016** also stores authentication and/or validation components **1044** configured to authenticate/validate specified EGM components and/or information, such as hardware components, software components, firmware components, peripheral device components, user input device components, information received from one or more user input devices, information stored in the at least one memory device **1016**, etc.

In certain embodiments, the peripheral devices **1022** include several device interfaces, such as: (1) at least one output device **1020** including at least one display device **1035**; (2) at least one input device **1030** (which may include

contact and/or non-contact interfaces); (3) at least one transponder **1054**; (4) at least one wireless communication component **1056**; (5) at least one wired/wireless power distribution component **1058**; (6) at least one sensor **1060**; (7) at least one data preservation component **1062**; (8) at least one motion/gesture analysis and interpretation component **1064**; (9) at least one motion detection component **1066**; (10) at least one portable power source **1068**; (11) at least one geolocation module **1076**; (12) at least one user identification module **1077**; (13) at least one user/device tracking module **1078**; and (14) at least one information filtering module **1079**.

The at least one output device **1020** includes at least one display device **1035** configured to display any game(s) displayed by the EGM and any suitable information associated with such game(s). In certain embodiments, the display devices are connected to or mounted on a housing of the EGM (described below). In various embodiments, the display devices serve as digital glass configured to advertise certain games or other aspects of the gaming establishment in which the EGM is located. In various embodiments, the EGM includes one or more of the following display devices: (a) a central display device; (b) a player tracking display configured to display various information regarding a user's player tracking status (as described below); (c) a secondary or upper display device in addition to the central display device and the player tracking display; (d) a credit display configured to display a current quantity of credits, amount of cash, account balance, or the equivalent; and (e) a bet display configured to display an amount wagered for one or more plays of one or more games. The example EGM **2000a** illustrated in FIG. **6A** includes a central display device **2116**, a player tracking display **2140**, a credit display **2120**, and a bet display **2122**. The example EGM **2000b** illustrated in FIG. **6B** includes a central display device **2116**, an upper display device **2118**, a player tracking display **2140**, a credit display **2120**, and a bet display **2122**.

In various embodiments, the display devices include, without limitation: a monitor, a television display, a plasma display, a liquid crystal display (LCD), a display based on light emitting diodes (LEDs), a display based on a plurality of organic light-emitting diodes (OLEDs), a display based on polymer light-emitting diodes (PLEDs), a display based on a plurality of surface-conduction electron-emitters (SEDs), a display including a projected and/or reflected image, or any other suitable electronic device or display mechanism. In certain embodiments, as described above, the display device includes a touchscreen with an associated touch-screen controller. The display devices may be of any suitable sizes, shapes, and configurations.

The display devices of the EGM are configured to display one or more game and/or non-game images, symbols, and indicia. In certain embodiments, the display devices of the EGM are configured to display any suitable visual representation or exhibition of the movement of objects; dynamic lighting; video images; images of people, characters, places, things, and faces of cards; and the like. In certain embodiments, the display devices of the EGM are configured to display one or more video reels, one or more video wheels, and/or one or more video dice. In other embodiments, certain of the displayed images, symbols, and indicia are in mechanical form. That is, in these embodiments, the display device includes any electromechanical device, such as one or more rotatable wheels, one or more reels, and/or one or more dice, configured to display at least one or a plurality of game or other suitable images, symbols, or indicia.

In various embodiments, the at least one output device **1020** includes a payout device. In these embodiments, after the EGM receives an actuation of a cashout device (described below), the EGM causes the payout device to provide a payment to the user. In one embodiment, the payout device is one or more of: (a) a ticket printer and dispenser configured to print and dispense a ticket or credit slip associated with a monetary value, wherein the ticket or credit slip may be redeemed for its monetary value via a cashier, a kiosk, or other suitable redemption system; (b) a bill dispenser configured to dispense paper currency; (c) a coin dispenser configured to dispense coins or tokens (such as into a coin payout tray); and (d) any suitable combination thereof. The example EGMs **2000a** and **2000b** illustrated in FIGS. **6A** and **6B** each include a ticket printer and dispenser **2136**.

In certain embodiments, rather than dispensing bills, coins, or a physical ticket having a monetary value to the user following receipt of an actuation of the cashout device, the payout device is configured to cause a payment to be provided to the user in the form of an electronic funds transfer, such as via a direct deposit into a bank account, a casino account, or a prepaid account of the user; via a transfer of funds onto an electronically recordable identification card or smart card of the user; or via sending a virtual ticket having a monetary value to an electronic device of the user.

While any credit balances, any wagers, any values, and any awards are described herein as amounts of monetary credits or currency, one or more of such credit balances, such wagers, such values, and such awards may be for non-monetary credits, promotional credits, of player tracking points or credits.

In certain embodiments, the at least one output device **1020** is a sound generating device controlled by one or more sound cards. In one such embodiment, the sound generating device includes one or more speakers or other sound generating hardware and/or software configured to generate sounds, such as by playing music for any games or by playing music for other modes of the EGM, such as an attract mode. The example EGMs **2000a** and **2000b** illustrated in FIGS. **6A** and **6B** each include a plurality of speakers **2150**. In another such embodiment, the EGM provides dynamic sounds coupled with attractive multimedia images displayed on one or more of the display devices to provide an audio-visual representation or to otherwise display full-motion video with sound to attract users to the EGM. In certain embodiments, the EGM displays a sequence of audio and/or visual attraction messages during idle periods to attract potential users to the EGM. The videos may be customized to provide any appropriate information.

The at least one input device **1030** may include any suitable device that enables an input signal to be produced and received by the at least one processor **1010** of the EGM.

In one embodiment, the at least one input device **1030** includes a payment device configured to communicate with the at least one processor of the EGM to fund the EGM. In certain embodiments, the payment device includes one or more of: (a) a bill acceptor into which paper money is inserted to fund the EGM; (b) a ticket acceptor into which a ticket or a voucher is inserted to fund the EGM; (c) a coin slot into which coins or tokens are inserted to fund the EGM; (d) a reader or a validator for credit cards, debit cards, or credit slips into which a credit card, debit card, or credit slip is inserted to fund the EGM; (e) a user identification card reader into which a user identification card is inserted to fund the EGM; or (f) any suitable combination thereof. The

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example EGMs **2000a** and **2000b** illustrated in FIGS. **6A** and **6B** each include a combined bill and ticket acceptor **2128** and a coin slot **2126**.

In one embodiment, the at least one input device **1030** includes a payment device configured to enable the EGM to be funded via an electronic funds transfer, such as a transfer of funds from a bank account. In another embodiment, the EGM includes a payment device configured to communicate with a mobile device of a user, such as a mobile phone, a radio frequency identification tag, or any other suitable wired or wireless device, to retrieve relevant information associated with that user to fund the EGM. When the EGM is funded, the at least one processor determines the amount of funds entered and displays the corresponding amount on a credit display or any other suitable display as described below.

In certain embodiments, the at least one input device **1030** includes at least one wagering or betting device. In various embodiments, the one or more wagering or betting devices are each: (1) a mechanical button supported by the housing of the EGM (such as a hard key or a programmable soft key), or (2) an icon displayed on a display device of the EGM (described below) that is actuatable via a touch screen of the EGM (described below) or via use of a suitable input device of the EGM (such as a mouse or a joystick). One such wagering or betting device is as a maximum wager or bet device that, when actuated, causes the EGM to place a maximum wager on a play of a game. Another such wagering or betting device is a repeat bet device that, when actuated, causes the EGM to place a wager that is equal to the previously placed wager on a play of a game. A further such wagering or betting device is a bet one device that, when actuated, causes the EGM to increase the wager by one credit. Generally, upon actuation of one of the wagering or betting devices, the quantity of credits displayed in a credit meter (described below) decreases by the amount of credits wagered, while the quantity of credits displayed in a bet display (described below) increases by the amount of credits wagered.

In various embodiments, the at least one input device **1030** includes at least one game play activation device. In various embodiments, the one or more game play initiation devices are each: (1) a mechanical button supported by the housing of the EGM (such as a hard key or a programmable soft key), or (2) an icon displayed on a display device of the EGM (described below) that is actuatable via a touch screen of the EGM (described below) or via use of a suitable input device of the EGM (such as a mouse or a joystick). After a user appropriately funds the EGM and places a wager, the EGM activates the game play activation device to enable the user to actuate the game play activation device to initiate a play of a game on the EGM (or another suitable sequence of events associated with the EGM). After the EGM receives an actuation of the game play activation device, the EGM initiates the play of the game. The example EGMs **2000a** and **2000b** illustrated in FIGS. **6A** and **6B** each include a game play activation device in the form of a game play initiation button **2132**. In other embodiments, the EGM begins game play automatically upon appropriate funding rather than upon utilization of the game play activation device.

In other embodiments, the at least one input device **1030** includes a cashout device. In various embodiments, the cashout device is: (1) a mechanical button supported by the housing of the EGM (such as a hard key or a programmable soft key), or (2) an icon displayed on a display device of the EGM (described below) that is actuatable via a touch screen

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of the EGM (described below) or via use of a suitable input device of the EGM (such as a mouse or a joystick). When the EGM receives an actuation of the cashout device from a user and the user has a positive (i.e., greater-than-zero) credit balance, the EGM initiates a payout associated with the user's credit balance. The example EGMs **2000a** and **2000b** illustrated in FIGS. **6A** and **6B** each include a cashout device in the form of a cashout button **2134**.

In various embodiments, the at least one input device **1030** includes a plurality of buttons that are programmable by the EGM operator to, when actuated, cause the EGM to perform particular functions. For instance, such buttons may be hard keys, programmable soft keys, or icons icon displayed on a display device of the EGM (described below) that are actuatable via a touch screen of the EGM (described below) or via use of a suitable input device of the EGM (such as a mouse or a joystick). The example EGMs **2000a** and **2000b** illustrated in FIGS. **6A** and **6B** each include a plurality of such buttons **2130**.

In certain embodiments, the at least one input device **1030** includes a touchscreen coupled to a touch-screen controller or other touch-sensitive display overlay to enable interaction with any images displayed on a display device (as described below). One such input device is a conventional touch-screen button panel. The touchscreen and the touch-screen controller are connected to a video controller. In these embodiments, signals are input to the EGM by touching the touch screen at the appropriate locations.

In embodiments including a player tracking system, as further described below, the at least one input device **1030** includes a card reader in communication with the at least one processor of the EGM. The example EGMs **2000a** and **2000b** illustrated in FIGS. **6A** and **6B** each include a card reader **2138**. The card reader is configured to read a user identification card inserted into the card reader.

The at least one wireless communication component **1056** includes one or more communication interfaces having different architectures and utilizing a variety of protocols, such as (but not limited to) 802.11 (WiFi); 802.15 (including Bluetooth™); 802.16 (WiMax); 802.22; cellular standards such as CDMA, CDMA2000, and WCDMA; Radio Frequency (e.g., RFID); infrared; and Near Field Magnetic communication protocols. The at least one wireless communication component **1056** transmits electrical, electro-magnetic, or optical signals that carry digital data streams or analog signals representing various types of information.

The at least one wired/wireless power distribution component **1058** includes components or devices that are configured to provide power to other devices. For example, in one embodiment, the at least one power distribution component **1058** includes a magnetic induction system that is configured to provide wireless power to one or more user input devices near the EGM. In one embodiment, a user input device docking region is provided, and includes a power distribution component that is configured to recharge a user input device without requiring metal-to-metal contact. In one embodiment, the at least one power distribution component **1058** is configured to distribute power to one or more internal components of the EGM, such as one or more rechargeable power sources (e.g., rechargeable batteries) located at the EGM.

In certain embodiments, the at least one sensor **1060** includes at least one of: optical sensors, pressure sensors, RF sensors, infrared sensors, image sensors, thermal sensors, and biometric sensors. The at least one sensor **1060** may be used for a variety of functions, such as: detecting movements and/or gestures of various objects within a predeter-

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mined proximity to the EGM; detecting the presence and/or identity of various persons (e.g., users, casino employees, etc.), devices (e.g., user input devices), and/or systems within a predetermined proximity to the EGM.

The at least one data preservation component **1062** is configured to detect or sense one or more events and/or conditions that, for example, may result in damage to the EGM and/or that may result in loss of information associated with the EGM. Additionally, the data preservation system **1062** may be operable to initiate one or more appropriate action(s) in response to the detection of such events/conditions.

The at least one motion/gesture analysis and interpretation component **1064** is configured to analyze and/or interpret information relating to detected user movements and/or gestures to determine appropriate user input information relating to the detected user movements and/or gestures. For example, in one embodiment, the at least one motion/gesture analysis and interpretation component **1064** is configured to perform one or more of the following functions: analyze the detected gross motion or gestures of a user; interpret the user's motion or gestures (e.g., in the context of a casino game being played) to identify instructions or input from the user; utilize the interpreted instructions/input to advance the game state; etc. In other embodiments, at least a portion of these additional functions may be implemented at a remote system or device.

The at least one portable power source **1068** enables the EGM to operate in a mobile environment. For example, in one embodiment, the EGM **300** includes one or more rechargeable batteries.

The at least one geolocation module **1076** is configured to acquire geolocation information from one or more remote sources and use the acquired geolocation information to determine information relating to a relative and/or absolute position of the EGM. For example, in one implementation, the at least one geolocation module **1076** is configured to receive GPS signal information for use in determining the position or location of the EGM. In another implementation, the at least one geolocation module **1076** is configured to receive multiple wireless signals from multiple remote devices (e.g., EGMs, servers, wireless access points, etc.) and use the signal information to compute position/location information relating to the position or location of the EGM.

The at least one user identification module **1077** is configured to determine the identity of the current user or current owner of the EGM. For example, in one embodiment, the current user is required to perform a login process at the EGM in order to access one or more features. Alternatively, the EGM is configured to automatically determine the identity of the current user based on one or more external signals, such as an RFID tag or badge worn by the current user and that provides a wireless signal to the EGM that is used to determine the identity of the current user. In at least one embodiment, various security features are incorporated into the EGM to prevent unauthorized users from accessing confidential or sensitive information.

The at least one information filtering module **1079** is configured to perform filtering (e.g., based on specified criteria) of selected information to be displayed at one or more displays **1035** of the EGM.

In various embodiments, the EGM includes a plurality of communication ports configured to enable the at least one processor of the EGM to communicate with and to operate with external peripherals, such as: accelerometers, arcade sticks, bar code readers, bill validators, biometric input devices, bonus devices, button panels, card readers, coin

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dispensers, coin hoppers, display screens or other displays or video sources, expansion buses, information panels, keypads, lights, mass storage devices, microphones, motion sensors, motors, printers, reels, SCSI ports, solenoids, speakers, thumbsticks, ticket readers, touch screens, trackballs, touchpads, wheels, and wireless communication devices.

As generally described above, in certain embodiments, such as the example EGMs **2000a** and **2000b** illustrated in FIGS. **6A** and **6B**, the EGM has a support structure, housing, or cabinet that provides support for a plurality of the input devices and the output devices of the EGM. Further, the EGM is configured such that a user may operate it while standing or sitting. In various embodiments, the EGM is positioned on a base or stand, or is configured as a pub-style tabletop game (not shown) that a user may operate typically while sitting. As illustrated by the different example EGMs **2000a** and **2000b** shown in FIGS. **6A** and **6B**, EGMs may have varying housing and display configurations.

In certain embodiments, the EGM is a device that has obtained approval from a regulatory gaming commission, and in other embodiments, the EGM is a device that has not obtained approval from a regulatory gaming commission.

The EGMs described above are merely three examples of different types of EGMs. Certain of these example EGMs may include one or more elements that may not be included in all gaming systems, and these example EGMs may not include one or more elements that are included in other gaming systems. For example, certain EGMs include a coin acceptor while others do not.

In various embodiments, an EGM may be implemented in one of a variety of different configurations. In various embodiments, the EGM may be implemented as one of: (a) a dedicated EGM in which computerized game programs executable by the EGM for controlling any primary or base games (referred to herein as "primary games") and/or any secondary or bonus games or other functions (referred to herein as "secondary games") displayed by the EGM are provided with the EGM before delivery to a gaming establishment or before being provided to a user; and (b) a changeable EGM in which computerized game programs executable by the EGM for controlling any primary games and/or secondary games displayed by the EGM are downloadable or otherwise transferred to the EGM through a data network or remote communication link; from a USB drive, flash memory card, or other suitable memory device; or in any other suitable manner after the EGM is physically located in a gaming establishment or after the EGM is provided to a user.

As generally explained above, in various embodiments in which the gaming system includes a server and a changeable EGM, the at least one memory device of the server stores different game programs and instructions executable by the at least one processor of the changeable EGM to control one or more primary games and/or secondary games displayed by the changeable EGM. More specifically, each such executable game program represents a different game or a different type of game that the at least one changeable EGM is configured to operate. In one example, certain of the game programs are executable by the changeable EGM to operate games having the same or substantially the same game play but different paytables. In different embodiments, each executable game program is associated with a primary game, a secondary game, or both. In certain embodiments, an executable game program is executable by the at least one processor of the at least one changeable EGM as a secondary game to be played simultaneously with a play of a primary

game (which may be downloaded to or otherwise stored on the at least one changeable EGM), or vice versa.

In operation of such embodiments, the server is configured to communicate one or more of the stored executable game programs to the at least one processor of the changeable EGM. In different embodiments, a stored executable game program is communicated or delivered to the at least one processor of the changeable EGM by: (a) embedding the executable game program in a device or a component (such as a microchip to be inserted into the changeable EGM); (b) writing the executable game program onto a disc or other media; or (c) uploading or streaming the executable game program over a data network (such as a dedicated data network). After the executable game program is communicated from the server to the changeable EGM, the at least one processor of the changeable EGM executes the executable game program to enable the primary game and/or the secondary game associated with that executable game program to be played using the display device(s) and/or the input device(s) of the changeable EGM. That is, when an executable game program is communicated to the at least one processor of the changeable EGM, the at least one processor of the changeable EGM changes the game or the type of game that may be played using the changeable EGM.

In certain embodiments, the gaming system randomly determines any game outcome(s) (such as a win outcome) and/or award(s) (such as a quantity of credits to award for the win outcome) for a play of a primary game and/or a play of a secondary game based on probability data. In certain such embodiments, this random determination is provided through utilization of an RNG, such as a true RNG or a pseudo RNG, or any other suitable randomization process. In one such embodiment, each game outcome or award is associated with a probability, and the gaming system generates the game outcome(s) and/or the award(s) to be provided based on the associated probabilities. In these embodiments, since the gaming system generates game outcomes and/or awards randomly or based on one or more probability calculations, there is no certainty that the gaming system will ever provide any specific game outcome and/or award.

In certain embodiments, the gaming system maintains one or more predetermined pools or sets of predetermined game outcomes and/or awards. In certain such embodiments, upon generation or receipt of a game outcome and/or award request, the gaming system independently selects one of the predetermined game outcomes and/or awards from the one or more pools or sets. The gaming system flags or marks the selected game outcome and/or award as used. Once a game outcome or an award is flagged as used, it is prevented from further selection from its respective pool or set; that is, the gaming system does not select that game outcome or award upon another game outcome and/or award request. The gaming system provides the selected game outcome and/or award.

In certain embodiments, the gaming system determines a predetermined game outcome and/or award based on the results of a bingo, keno, or lottery game. In certain such embodiments, the gaming system utilizes one or more bingo, keno, or lottery games to determine the predetermined game outcome and/or award provided for a primary game and/or a secondary game. The gaming system is provided or associated with a bingo card. Each bingo card consists of a matrix or array of elements, wherein each element is designated with separate indicia. After a bingo card is provided, the gaming system randomly selects or draws a plurality of the elements. As each element is selected, a determination is made as to whether the selected element is present on the

bingo card. If the selected element is present on the bingo card, that selected element on the provided bingo card is marked or flagged. This process of selecting elements and marking any selected elements on the provided bingo cards continues until one or more predetermined patterns are marked on one or more of the provided bingo cards. After one or more predetermined patterns are marked on one or more of the provided bingo cards, game outcome and/or award is determined based, at least in part, on the selected elements on the provided bingo cards.

In certain embodiments in which the gaming system includes a server and an EGM, the EGM is configured to communicate with the server for monitoring purposes only. In such embodiments, the EGM determines the game outcome(s) and/or award(s) to be provided in any of the manners described above, and the server monitors the activities and events occurring on the EGM. In one such embodiment, the gaming system includes a real-time or online accounting and gaming information system configured to communicate with the server. In this embodiment, the accounting and gaming information system includes: (a) a user database configured to store user profiles, (b) a player tracking module configured to track users (as described below), and (c) a credit system configured to provide automated transactions.

As noted above, in various embodiments, the gaming system includes one or more executable game programs executable by at least one processor of the gaming system to provide one or more primary games and one or more secondary games. The primary game(s) and the secondary game(s) may comprise any suitable games and/or wagering games, such as, but not limited to electro-mechanical or video slot or spinning reel type games; video card games such as video draw poker, multi-hand video draw poker, other video poker games, video blackjack games, and video baccarat games; video keno games; video bingo games; and video selection games.

In certain embodiments in which the primary game is a slot or spinning reel type game, the gaming system includes one or more reels in either an electromechanical form with mechanical rotating reels or in a video form with simulated reels and movement thereof. Each reel displays a plurality of indicia or symbols, such as bells, hearts, fruits, numbers, letters, bars, or other images that typically correspond to a theme associated with the gaming system. In certain such embodiments, the gaming system includes one or more paylines associated with the reels. The example EGM **2000b** shown in FIG. 6B includes a payline **2152** and a plurality of reels **2154**. In certain embodiments, one or more of the reels are independent reels or unisymbol reels. In such embodiments, each independent reel generates and displays one symbol.

In various embodiments, one or more of the paylines is horizontal, vertical, circular, diagonal, angled, or any suitable combination thereof. In other embodiments, each of one or more of the paylines is associated with a plurality of adjacent symbol display areas on a requisite number of adjacent reels. In one such embodiment, one or more paylines are formed between at least two symbol display areas that are adjacent to each other by either sharing a common side or sharing a common corner (i.e., such paylines are connected paylines). The gaming system enables a wager to be placed on one or more of such paylines to activate such paylines. In other embodiments in which one or more paylines are formed between at least two adjacent symbol display areas, the gaming system enables a wager to be

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placed on a plurality of symbol display areas, which activates those symbol display areas.

In various embodiments, the gaming system provides one or more awards after a spin of the reels when specified types and/or configurations of the indicia or symbols on the reels occur on an active payline or otherwise occur in a winning pattern, occur on the requisite number of adjacent reels, and/or occur in a scatter pay arrangement.

In certain embodiments, the gaming system employs a ways to win award determination. In these embodiments, any outcome to be provided is determined based on a number of associated symbols that are generated in active symbol display areas on the requisite number of adjacent reels (i.e., not on paylines passing through any displayed winning symbol combinations). If a winning symbol combination is generated on the reels, one award for that occurrence of the generated winning symbol combination is provided.

In various embodiments, the gaming system includes a progressive award. Typically, a progressive award includes an initial amount and an additional amount funded through a portion of each wager placed to initiate a play of a primary game. When one or more triggering events occurs, the gaming system provides at least a portion of the progressive award. After the gaming system provides the progressive award, an amount of the progressive award is reset to the initial amount and a portion of each subsequent wager is allocated to the next progressive award.

As generally noted above, in addition to providing winning credits or other awards for one or more plays of the primary game(s), in various embodiments the gaming system provides credits or other awards for one or more plays of one or more secondary games. The secondary game typically enables an award to be obtained addition to any award obtained through play of the primary game(s). The secondary game(s) typically produces a higher level of user excitement than the primary game(s) because the secondary game(s) provides a greater expectation of winning than the primary game(s) and is accompanied with more attractive or unusual features than the primary game(s). The secondary game(s) may be any type of suitable game, either similar to or completely different from the primary game.

In various embodiments, the gaming system automatically provides or initiates the secondary game upon the occurrence of a triggering event or the satisfaction of a qualifying condition. In other embodiments, the gaming system initiates the secondary game upon the occurrence of the triggering event or the satisfaction of the qualifying condition and upon receipt of an initiation input. In certain embodiments, the triggering event or qualifying condition is a selected outcome in the primary game(s) or a particular arrangement of one or more indicia on a display device for a play of the primary game(s), such as a "BONUS" symbol appearing on three adjacent reels along a payline following a spin of the reels for a play of the primary game. In other embodiments, the triggering event or qualifying condition occurs based on a certain amount of game play (such as number of games, number of credits, amount of time) being exceeded, or based on a specified number of points being earned during game play. Any suitable triggering event or qualifying condition or any suitable combination of a plurality of different triggering events or qualifying conditions may be employed.

In other embodiments, at least one processor of the gaming system randomly determines when to provide one or more plays of one or more secondary games. In one such embodiment, no apparent reason is provided for providing

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the secondary game. In this embodiment, qualifying for a secondary game is not triggered by the occurrence of an event in any primary game or based specifically on any of the plays of any primary game. That is, qualification is provided without any explanation or, alternatively, with a simple explanation. In another such embodiment, the gaming system determines qualification for a secondary game at least partially based on a game triggered or symbol triggered event, such as at least partially based on play of a primary game.

In various embodiments, after qualification for a secondary game has been determined, the secondary game participation may be enhanced through continued play on the primary game. Thus, in certain embodiments, for each secondary game qualifying event, such as a secondary game symbol, that is obtained, a given number of secondary game wagering points or credits is accumulated in a "secondary game meter" configured to accrue the secondary game wagering credits or entries toward eventual participation in the secondary game. In one such embodiment, the occurrence of multiple such secondary game qualifying events in the primary game results in an arithmetic or exponential increase in the number of secondary game wagering credits awarded. In another such embodiment, any extra secondary game wagering credits may be redeemed during the secondary game to extend play of the secondary game.

In certain embodiments, no separate entry fee or buy-in for the secondary game is required. That is, entry into the secondary game cannot be purchased; rather, in these embodiments' entry must be won or earned through play of the primary game, thereby encouraging play of the primary game. In other embodiments, qualification for the secondary game is accomplished through a simple "buy-in." For example, qualification through other specified activities is unsuccessful, payment of a fee or placement of an additional wager "buys-in" to the secondary game. In certain embodiments, a separate side wager must be placed on the secondary game, or a wager of a designated amount must be placed on the primary game to enable qualification for the secondary game. In these embodiments, the secondary game triggering event must occur and the side wager (or designated primary game wager amount) must have been placed for the secondary game to trigger.

In various embodiments in which the gaming system includes a plurality of EGMs, the EGMs are configured to communicate with one another to provide a group gaming environment. In certain such embodiments, the EGMs enable users of those EGMs to work in conjunction with one another, such as by enabling the users to play together as a team or group, to win one or more awards. In other such embodiments, the EGMs enable users of those EGMs to compete against one another for one or more awards. In one such embodiment, the EGMs enable the users of those EGMs to participate in one or more gaming tournaments for one or more awards.

In various embodiments, the gaming system includes one or more player tracking systems. Such player tracking systems enable operators of the gaming system (such as casinos or other gaming establishments) to recognize the value of customer loyalty by identifying frequent customers and rewarding them for their patronage. Such a player tracking system is configured to track a user's gaming activity. In one such embodiment, the player tracking system does so through the use of player tracking cards. In this embodiment, a user is issued a user identification card that has an encoded user identification number that uniquely identifies the user. When the user's playing tracking card is

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inserted into a card reader of the gaming system to begin a gaming session, the card reader reads the user identification number off the player tracking card to identify the user. The gaming system timely tracks any suitable information or data relating to the identified user's gaming session. The gaming system also timely tracks when the player tracking card is removed to conclude play for that gaming session. In another embodiment, rather than requiring insertion of a player tracking card into the card reader, the gaming system utilizes one or more portable devices, such as a mobile phone, a radio frequency identification tag, or any other suitable wireless device, to track when a gaming session begins and ends. In another embodiment, the gaming system utilizes any suitable biometric technology or ticket technology to track when a gaming session begins and ends.

In such embodiments, during one or more gaming sessions, the gaming system tracks any suitable information or data, such as any amounts wagered, average wager amounts, and/or the time at which these wagers are placed. In different embodiments, for one or more users, the player tracking system includes the user's account number, the user's card number, the user's first name, the user's surname, the user's preferred name, the user's player tracking ranking, any promotion status associated with the user's player tracking card, the user's address, the user's birthday, the user's anniversary, the user's recent gaming sessions, or any other suitable data. In various embodiments, such tracked information and/or any suitable feature associated with the player tracking system is displayed on a player tracking display. In various embodiments, such tracked information and/or any suitable feature associated with the player tracking system is displayed via one or more service windows that are displayed on the central display device and/or the upper display device.

In various embodiments, the gaming system includes one or more servers configured to communicate with a personal gaming device—such as a smartphone, a tablet computer, a desktop computer, or a laptop computer—to enable web-based game play using the personal gaming device. In various embodiments, the user must first access a gaming website via an Internet browser of the personal gaming device or execute an application (commonly called an “app”) installed on the personal gaming device before the user can use the personal gaming device to participate in web-based game play. In certain embodiments, the one or more servers and the personal gaming device operate in a thin-client environment. In these embodiments, the personal gaming device receives inputs via one or more input devices (such as a touch screen and/or physical buttons), the personal gaming device sends the received inputs to the one or more servers, the one or more servers make various determinations based on the inputs and determine content to be displayed (such as a randomly determined game outcome and corresponding award), the one or more servers send the content to the personal gaming device, and the personal gaming device displays the content.

In certain such embodiments, the one or more servers must identify the user before enabling game play on the personal gaming device (or, in some embodiments, before enabling monetary wager-based game play on the personal gaming device). In these embodiments, the user must identify herself to the one or more servers, such as by inputting the user's unique username and password combination, providing an input to a biometric sensor (e.g., a fingerprint sensor, a retinal sensor, a voice sensor, or a facial-recognition sensor), or providing any other suitable information.

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Once identified, the one or more servers enable the user to establish an account balance from which the user can draw credits usable to wager on plays of a game. In certain embodiments, the one or more servers enable the user to initiate an electronic funds transfer to transfer funds from a bank account to the user's account balance. In other embodiments, the one or more servers enable the user to make a payment using the user's credit card, debit card, or other suitable device to add money to the user's account balance. In other embodiments, the one or more servers enable the user to add money to the user's account balance via a peer-to-peer type application, such as PayPal or Venmo. The one or more servers also enable the user to cash out the user's account balance (or part of it) in any suitable manner, such as via an electronic funds transfer, by initiating creation of a paper check that is mailed to the user, or by initiating printing of a voucher at a kiosk in a gaming establishment.

In certain embodiments, the one or more servers include a payment server that handles establishing and cashing out users' account balances and a separate game server configured to determine the outcome and any associated award for a play of a game. In these embodiments, the game server is configured to communicate with the personal gaming device and the payment device, and the personal gaming device and the payment device are not configured to directly communicate with one another. In these embodiments, when the game server receives data representing a request to start a play of a game at a desired wager, the game server sends data representing the desired wager to the payment server. The payment server determines whether the user's account balance can cover the desired wager (i.e., includes a monetary balance at least equal to the desired wager).

If the payment server determines that the user's account balance cannot cover the desired wager, the payment server notifies the game server, which then instructs the personal gaming device to display a suitable notification to the user that the user's account balance is too low to place the desired wager. If the payment server determines that the user's account balance can cover the desired wager, the payment server deducts the desired wager from the account balance and notifies the game server. The game server then determines an outcome and any associated award for the play of the game. The game server notifies the payment server of any nonzero award, and the payment server increases the user's account balance by the nonzero award. The game server sends data representing the outcome and any award to the personal gaming device, which displays the outcome and any award.

In certain embodiments, the one or more servers enable web-based game play using a personal gaming device only if the personal gaming device satisfies one or more jurisdictional requirements. In one embodiment, the one or more servers enable web-based game play using the personal gaming device only if the personal gaming device is located within a designated geographic area (such as within certain state or county lines or within the boundaries of a gaming establishment). In this embodiment, the geolocation module of the personal gaming device determines the location of the personal gaming device and sends the location to the one or more servers, which determine whether the personal gaming device is located within the designated geographic area. In various embodiments, the one or more servers enable non-monetary wager-based game play if the personal gaming device is located outside of the designated geographic area.

In various embodiments, the gaming system includes an EGM configured to communicate with a personal gaming device—such as a smartphone, a tablet computer, a desktop

computer, or a laptop computer—to enable tethered mobile game play using the personal gaming device. Generally, in these embodiments, the EGM establishes communication with the personal gaming device and enables the user to play games on the EGM remotely via the personal gaming device. In certain embodiments, the gaming system includes a geo-fence system that enables tethered game play within a particular geographic area but not outside of that geographic area.

In certain embodiments, the gaming system is configured to communicate with a social network server that hosts or partially hosts a social networking website via a data network (such as the Internet) to integrate a user's gaming experience with the user's social networking account. This enables the gaming system to send certain information to the social network server that the social network server can use to create content (such as text, an image, and/or a video) and post it to the user's wall, newsfeed, or similar area of the social networking website accessible by the user's connections (and in certain cases the public) such that the user's connections can view that information. This also enables the gaming system to receive certain information from the social network server, such as the user's likes or dislikes or the user's list of connections. In certain embodiments, the gaming system enables the user to link the user's user account to the user's social networking account(s). This enables the gaming system to, once it identifies the user and initiates a gaming session (such as via the user logging in to a website (or an application) on the user's personal gaming device or via the user inserting the user's player tracking card into an EGM), link that gaming session to the user's social networking account(s). In other embodiments, the gaming system enables the user to link the user's social networking account(s) to individual gaming sessions when desired by providing the required login information.

For instance, in one embodiment, if a user wins a particular award (e.g., a progressive award or a jackpot award) or an award that exceeds a certain threshold (e.g., an award exceeding \$1,000), the gaming system sends information about the award to the social network server to enable the server to create associated content (such as a screenshot of the outcome and associated award) and to post that content to the user's wall (or other suitable area) of the social networking website for the user's connections to see (and to entice them to play). In another embodiment, if a user joins a multiplayer game and there is another seat available, the gaming system sends that information to the social network server to enable the server to create associated content (such as text indicating a vacancy for that particular game) and to post that content to the user's wall (or other suitable area) of the social networking website for the user's connections to see (and to entice them to fill the vacancy). In another embodiment, if the user consents, the gaming system sends advertisement information or offer information to the social network server to enable the social network server to create associated content (such as text or an image reflecting an advertisement and/or an offer) and to post that content to the user's wall (or other suitable area) of the social networking website for the user's connections to see. In another embodiment, the gaming system enables the user to recommend a game to the user's connections by posting a recommendation to the user's wall (or other suitable area) of the social networking website.

Certain of the gaming systems described herein, such as EGMs located in a casino or another gaming establishment, include certain components and/or are configured to operate in certain manners that differentiate these systems from

general purpose computing devices, i.e., certain personal gaming devices such as desktop computers and laptop computers.

For instance, EGMs are highly regulated to ensure fairness, and, in many cases, EGMs are configured to award monetary awards up to multiple millions of dollars. To satisfy security and regulatory requirements in a gaming environment, hardware and/or software architectures are implemented in EGMs that differ significantly from those of general purpose computing devices. For purposes of illustration, a description of EGMs relative to general purpose computing devices and some examples of these additional (or different) hardware and/or software architectures found in EGMs are described below.

At first glance, one might think that adapting general purpose computing device technologies to the gaming industry and EGMs would be a simple proposition because both general purpose computing devices and EGMs employ processors that control a variety of devices. However, due to at least: (1) the regulatory requirements placed on EGMs, (2) the harsh environment in which EGMs operate, (3) security requirements, and (4) fault tolerance requirements, adapting general purpose computing device technologies to EGMs can be quite difficult. Further, techniques and methods for solving a problem in the general purpose computing device industry, such as device compatibility and connectivity issues, might not be adequate in the gaming industry. For instance, a fault or a weakness tolerated in a general purpose computing device, such as security holes in software or frequent crashes, is not tolerated in an EGM because in an EGM these faults can lead to a direct loss of funds from the EGM, such as stolen cash or loss of revenue when the EGM is not operating properly or when the random outcome determination is manipulated.

Certain differences between general purpose computing devices and EGMs are described below. A first difference between EGMs and general purpose computing devices is that EGMs are state-based systems. A state-based system stores and maintains its current state in a non-volatile memory such that, in the event of a power failure or other malfunction, the state-based system can return to that state when the power is restored, or the malfunction is remedied. For instance, for a state-based EGM, if the EGM displays an award for a game of chance but the power to the EGM fails before the EGM provides the award to the user, the EGM stores the pre-power failure state in a non-volatile memory, returns to that state upon restoration of power, and provides the award to the user. This requirement affects the software and hardware design on EGMs. General purpose computing devices are not state-based machines, and a majority of data is usually lost when a malfunction occurs on a general purpose computing device.

A second difference between EGMs and general purpose computing devices is that, for regulatory purposes, the software on the EGM utilized to operate the EGM has been designed to be static and monolithic to prevent cheating by the operator of the EGM. For instance, one solution that has been employed in the gaming industry to prevent cheating and to satisfy regulatory requirements has been to manufacture an EGM that can use a proprietary processor running instructions to provide the game of chance from an EPROM or other form of non-volatile memory. The coding instructions on the EPROM are static (non-changeable) and must be approved by a gaming regulators in a particular jurisdiction and installed in the presence of a person representing the gaming jurisdiction. Any changes to any part of the software required to generate the game of chance, such as adding a

new device driver used to operate a device during generation of the game of chance, can require burning a new EPROM approved by the gaming jurisdiction and reinstalling the new EPROM on the EGM in the presence of a gaming regulator. Regardless of whether the EPROM solution is used, to gain approval in most gaming jurisdictions, an EGM must demonstrate sufficient safeguards that prevent an operator or a user of an EGM from manipulating the EGM's hardware and software in a manner that gives him an unfair, and in some cases illegal, advantage.

A third difference between EGMs and general purpose computing devices is authentication—EGMs storing code are configured to authenticate the code to determine if the code is unaltered before executing the code. If the code has been altered, the EGM prevents the code from being executed. The code authentication requirements in the gaming industry affect both hardware and software designs on EGMs. Certain EGMs use hash functions to authenticate code. For instance, one EGM stores game program code, a hash function, and an authentication hash (which may be encrypted). Before executing the game program code, the EGM hashes the game program code using the hash function to obtain a result hash and compares the result hash to the authentication hash. If the result hash matches the authentication hash, the EGM determines that the game program code is valid and executes the game program code. If the result hash does not match the authentication hash, the EGM determines that the game program code has been altered (i.e., may have been tampered with) and prevents execution of the game program code.

A fourth difference between EGMs and general purpose computing devices is that EGMs have unique peripheral device requirements that differ from those of a general purpose computing device, such as peripheral device security requirements not usually addressed by general purpose computing devices. For instance, monetary devices, such as coin dispensers, bill validators, and ticket printers and computing devices that are used to govern the input and output of cash or other items having monetary value (such as tickets) to and from an EGM have security requirements that are not typically addressed in general purpose computing devices. Therefore, many general purpose computing device techniques and methods developed to facilitate device connectivity and device compatibility do not address the emphasis placed on security in the gaming industry.

To address some of the issues described above, a number of hardware/software components and architectures are utilized in EGMs that are not typically found in general purpose computing devices. These hardware/software components and architectures, as described below in more detail, include but are not limited to watchdog timers, voltage monitoring systems, state-based software architecture and supporting hardware, specialized communication interfaces, security monitoring, and trusted memory.

Certain EGMs use a watchdog timer to provide a software failure detection mechanism. In a normally operating EGM, the operating software periodically accesses control registers in the watchdog timer subsystem to "re-trigger" the watchdog. Should the operating software fail to access the control registers within a preset timeframe, the watchdog timer will timeout and generate a system reset. Typical watchdog timer circuits include a loadable timeout counter register to enable the operating software to set the timeout interval within a certain range of time. A differentiating feature of some circuits is that the operating software cannot completely disable the function of the watchdog timer. In

other words, the watchdog timer always functions from the time power is applied to the board.

Certain EGMs use several power supply voltages to operate portions of the computer circuitry. These can be generated in a central power supply or locally on the computer board. If any of these voltages falls out of the tolerance limits of the circuitry they power, unpredictable operation of the EGM may result. Though most modern general purpose computing devices include voltage monitoring circuitry, these types of circuits only report voltage status to the operating software. Out of tolerance voltages can cause software malfunction, creating a potential uncontrolled condition in the general purpose computing device. Certain EGMs have power supplies with relatively tighter voltage margins than that required by the operating circuitry. In addition, the voltage monitoring circuitry implemented in certain EGMs typically has two thresholds of control. The first threshold generates a software event that can be detected by the operating software and an error condition then generated. This threshold is triggered when a power supply voltage falls out of the tolerance range of the power supply but is still within the operating range of the circuitry. The second threshold is set when a power supply voltage falls out of the operating tolerance of the circuitry. In this case, the circuitry generates a reset, halting operation of the EGM.

As described above, certain EGMs are state-based machines. Different functions of the game provided by the EGM (e.g., bet, play, result, points in the graphical presentation, etc.) may be defined as a state. When the EGM moves a game from one state to another, the EGM stores critical data regarding the game software in a custom non-volatile memory subsystem. This ensures that the user's wager and credits are preserved and to minimize potential disputes in the event of a malfunction on the EGM. In general, the EGM does not advance from a first state to a second state until critical information that enables the first state to be reconstructed has been stored. This feature enables the EGM to recover operation to the current state of play in the event of a malfunction, loss of power, etc. that occurred just before the malfunction. In at least one embodiment, the EGM is configured to store such critical information using atomic transactions.

Generally, an atomic operation in computer science refers to a set of operations that can be combined so that they appear to the rest of the system to be a single operation with only two possible outcomes: success or failure. As related to data storage, an atomic transaction may be characterized as series of database operations which either all occur, or all do not occur. A guarantee of atomicity prevents updates to the database occurring only partially, which can result in data corruption.

To ensure the success of atomic transactions relating to critical information to be stored in the EGM memory before a failure event (e.g., malfunction, loss of power, etc.), memory that includes one or more of the following criteria be used: direct memory access capability; data read/write capability which meets or exceeds minimum read/write access characteristics (such as at least 5.08 Mbytes/sec (Read) and/or at least 38.0 Mbytes/sec (Write)). Memory devices that meet or exceed the above criteria may be referred to as "fault-tolerant" memory devices.

Typically, battery-backed RAM devices may be configured to function as fault-tolerant devices according to the above criteria, whereas flash RAM and/or disk drive memory are typically not configurable to function as fault-tolerant devices according to the above criteria. Accordingly,

battery-backed RAM devices are typically used to preserve EGM critical data, although other types of non-volatile memory devices may be employed. These memory devices are typically not used in typical general purpose computing devices.

Thus, in at least one embodiment, the EGM is configured to store critical information in fault-tolerant memory (e.g., battery-backed RAM devices) using atomic transactions. Further, in at least one embodiment, the fault-tolerant memory is able to successfully complete all desired atomic transactions (e.g., relating to the storage of EGM critical information) within a time period of 200 milliseconds or less. In at least one embodiment, the time period of 200 milliseconds represents a maximum amount of time for which sufficient power may be available to the various EGM components after a power outage event has occurred at the EGM.

As described previously, the EGM may not advance from a first state to a second state until critical information that enables the first state to be reconstructed has been atomically stored. After the state of the EGM is restored during the play of a game of chance, game play may resume, and the game may be completed in a manner that is no different than if the malfunction had not occurred. Thus, for example, when a malfunction occurs during a game of chance, the EGM may be restored to a state in the game of chance just before when the malfunction occurred. The restored state may include metering information and graphical information that was displayed on the EGM in the state before the malfunction. For example, when the malfunction occurs during the play of a card game after the cards have been dealt, the EGM may be restored with the cards that were previously displayed as part of the card game. As another example, a bonus game may be triggered during the play of a game of chance in which a user is required to make a number of selections on a video display screen. When a malfunction has occurred after the user has made one or more selections, the EGM may be restored to a state that shows the graphical presentation just before the malfunction including an indication of selections that have already been made by the user. In general, the EGM may be restored to any state in a plurality of states that occur in the game of chance that occurs while the game of chance is played or to states that occur between the play of a game of chance.

Game history information regarding previous games played such as an amount wagered, the outcome of the game, and the like may also be stored in a non-volatile memory device. The information stored in the non-volatile memory may be detailed enough to reconstruct a portion of the graphical presentation that was previously presented on the EGM and the state of the EGM (e.g., credits) at the time the game of chance was played. The game history information may be utilized in the event of a dispute. For example, a user may decide that in a previous game of chance that they did not receive credit for an award that they believed they won. The game history information may be used to reconstruct the state of the EGM before, during, and/or after the disputed game to demonstrate whether the user was correct or not in the user's assertion.

Another feature of EGMs is that they often include unique interfaces, including serial interfaces, to connect to specific subsystems internal and external to the EGM. The serial devices may have electrical interface requirements that differ from the "standard" EIA serial interfaces provided by general purpose computing devices. These interfaces may include, for example, Fiber Optic Serial, optically coupled serial interfaces, current loop style serial interfaces, etc. In

addition, to conserve serial interfaces internally in the EGM, serial devices may be connected in a shared, daisy-chain fashion in which multiple peripheral devices are connected to a single serial channel.

The serial interfaces may be used to transmit information using communication protocols that are unique to the gaming industry. For example, IGT's Netplex is a proprietary communication protocol used for serial communication between EGMs. As another example, SAS is a communication protocol used to transmit information, such as metering information, from an EGM to a remote device. Often SAS is used in conjunction with a player tracking system.

Certain EGMs may alternatively be treated as peripheral devices to a casino communication controller and connected in a shared daisy chain fashion to a single serial interface. In both cases, the peripheral devices are assigned device addresses. If so, the serial controller circuitry must implement a method to generate or detect unique device addresses. General purpose computing device serial ports are not able to do this.

Security monitoring circuits detect intrusion into an EGM by monitoring security switches attached to access doors in the EGM cabinet. Access violations result in suspension of game play and can trigger additional security operations to preserve the current state of game play. These circuits also function when power is off by use of a battery backup. In power-off operation, these circuits continue to monitor the access doors of the EGM. When power is restored, the EGM can determine whether any security violations occurred while power was off, e.g., via software for reading status registers. This can trigger event log entries and further data authentication operations by the EGM software.

Trusted memory devices and/or trusted memory sources are included in an EGM to ensure the authenticity of the software that may be stored on less secure memory subsystems, such as mass storage devices. Trusted memory devices and controlling circuitry are typically designed to not enable modification of the code and data stored in the memory device while the memory device is installed in the EGM. The code and data stored in these devices may include authentication algorithms, random number generators, authentication keys, operating system kernels, etc. The purpose of these trusted memory devices is to provide gaming regulatory authorities a root trusted authority within the computing environment of the EGM that can be tracked and verified as original. This may be accomplished via removal of the trusted memory device from the EGM computer and verification of the secure memory device contents is a separate third party verification device. Once the trusted memory device is verified as authentic and based on the approval of the verification algorithms included in the trusted device, the EGM is enabled to verify the authenticity of additional code and data that may be located in the gaming computer assembly, such as code and data stored on hard disk drives.

In at least one embodiment, at least a portion of the trusted memory devices/sources may correspond to memory that cannot easily be altered (e.g., "unalterable memory") such as EPROMS, PROMS, Bios, Extended Bios, and/or other memory sources that are able to be configured, verified, and/or authenticated (e.g., for authenticity) in a secure and controlled manner.

According to one embodiment, when a trusted information source is in communication with a remote device via a network, the remote device may employ a verification scheme to verify the identity of the trusted information source. For example, the trusted information source and the

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remote device may exchange information using public and private encryption keys to verify each other's identities. In another embodiment, the remote device and the trusted information source may engage in methods using zero knowledge proofs to authenticate each of their respective identities.

EGMs storing trusted information may utilize apparatuses or methods to detect and prevent tampering. For instance, trusted information stored in a trusted memory device may be encrypted to prevent its misuse. In addition, the trusted memory device may be secured behind a locked door. Further, one or more sensors may be coupled to the memory device to detect tampering with the memory device and provide some record of the tampering. In yet another example, the memory device storing trusted information might be designed to detect tampering attempts and clear or erase itself when an attempt at tampering has been detected.

Mass storage devices used in a general purpose computing devices typically enable code and data to be read from and written to the mass storage device. In a gaming environment, modification of the gaming code stored on a mass storage device is strictly controlled and would only be enabled under specific maintenance type events with electronic and physical enablers required. Though this level of security could be provided by software, EGMs that include mass storage devices include hardware level mass storage data protection circuitry that operates at the circuit level to monitor attempts to modify data on the mass storage device and will generate both software and hardware error triggers should a data modification be attempted without the proper electronic and physical enablers being present.

It should be appreciated that the terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting of the disclosure. For example, the singular forms "a", "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. In another example, the terms "including" and "comprising" and variations thereof, when used in this specification, specify the presence of states features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof. Additionally, a listing of items does not imply that any or all of the items are mutually exclusive nor does a listing of items imply that any or all of the items are collectively exhaustive of anything or in a particular order, unless expressly specified otherwise. Moreover, as used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items. It should be further appreciated that headings of sections provided in this document and the title are for convenience only and are not to be taken as limiting the disclosure in any way. Furthermore, unless expressly specified otherwise, devices that are in communication with each other need not be in continuous communication with each other and may communicate directly or indirectly through one or more intermediaries.

Various changes and modifications to the present embodiments described herein will be apparent to those skilled in the art. For example, a description of an embodiment with several components in communication with each other does not imply that all such components are required, or that each of the disclosed components must communicate with every other component. On the contrary a variety of optional components are described to illustrate the wide variety of possible embodiments of the present disclosure. As such, these changes and modifications can be made without departing from the spirit and scope of the present subject

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matter and without diminishing its intended technical scope. It is therefore intended that such changes and modifications be covered by the appended claims.

The invention claimed is:

1. A gaming system comprising:

a processor; and

a memory device that stores a plurality of instructions that, when executed by the processor in association with a play of a game, cause the processor to:

cause a display, by a display device, of symbol displays comprising different symbols;

cause a display, by the display device, of an indication of selectable symbol display positions associated with the symbol displays;

cause a display, by the display device, of an indication of different probabilities of winning based on how many symbol display positions are selected as active symbol display positions;

cause a display, by the display device, of an indication of a selected plurality of the symbol display positions as being the active symbol display positions for the play of the game;

cause a display, by the display device, of a plurality of the different symbols on the symbol displays displayed at the plurality of the active symbol display positions; and

cause a display, by the display device, of an indication of a first quantity of same symbols at a plurality of the active symbol display positions,

wherein whether the first quantity of same symbols is a winning symbol combination depends on the symbol display positions selected as the active symbol display positions,

wherein if a first quantity of symbol display positions are the active symbol display positions, the first quantity of same symbols at the active symbol display positions forms the winning symbol combination, and

wherein the first quantity of same symbols at the active symbol display positions do not by themselves form the winning symbol combination if a second quantity of symbols display positions are selected as the symbol display positions and the second quantity is greater than the first quantity.

2. The gaming system of claim 1, wherein the active symbol display positions are less than all of the symbol display positions.

3. The gaming system of claim 1, wherein the display, by the display device, of the indication of selectable symbol display positions comprises an indication of different selectable columns of the symbol display positions as active columns of symbol display positions.

4. The gaming system of claim 3, wherein the plurality of instructions, when executed by the processor, cause the processor to cause a display, by the display device, of an indication of different probabilities of winning based on how many columns of symbol display positions are selected as active columns of symbol display positions.

5. The gaming system of claim 1, wherein the display, by the display device, of the indication of selectable symbol display positions comprises an indication of different selectable rows of the symbol display positions as active rows of symbol display positions.

6. The gaming system of claim 5, wherein the plurality of instructions, when executed by the processor, cause the processor to cause a display, by the display device, of an indication of different probabilities of winning based on how

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many rows of symbol display positions are selected as active rows of symbol display positions.

7. A gaming system comprising:

a processor; and

a memory device that stores a plurality of instructions that, when executed by the processor, cause the processor to:

cause a display, by a display device, of symbol displays comprising different symbols;

cause a display, by the display device, of an indication of selectable symbol display positions associated with the symbol display;

cause a display, by the display device, of an indication of different probabilities of winning based on how many symbol display positions are selected as active symbol display positions;

for a first play of a game:

cause a display, by the display device, of an indication of all of the symbol display positions as being first active symbol display positions for the first play of the game;

cause a display, by the display device, of a first plurality of the different symbols on the symbol displays displayed at the first active symbol display positions; and

cause a display, by the display device, of an indication of a first quantity of same symbols at the first active symbol display positions, wherein whether the first quantity of same symbols is a winning symbol combination depends on the symbol display positions selected as the first active symbol display positions, wherein the first quantity of same symbols at the first active symbol display positions does not form the winning symbol combination; and

for a second play of the game:

cause a display, by the display device, of an indication of less than all of the symbol display positions as being second active symbol display positions for the second play of the game;

cause a display, by the display device, of a second plurality of the different symbols on the symbol displays displayed at the second active symbol display positions; and

cause a display, by the display device, of an indication the first quantity of same symbols at the second active symbol display positions, wherein whether the first quantity of same symbols is the winning symbol combination depends on the symbol display positions selected as the second active symbol display positions, wherein the first quantity of same symbols at the second active symbol display positions form the winning symbol combination.

8. The gaming system of claim 7, wherein the display, by the display device, of the indication of selectable symbol display positions comprises an indication of different selectable columns of the symbol display positions as active columns of symbol display positions.

9. The gaming system of claim 8, wherein the plurality of instructions, when executed by the processor, cause the processor to cause a display, by the display device, of an indication of different probabilities of winning based on how many columns of symbol display positions are selected as active columns of symbol display positions.

10. The gaming system of claim 9, wherein the display, by the display device, of the indication of selectable symbol

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display positions comprises an indication of different selectable rows of the symbol display positions as active rows of symbol display positions.

11. The gaming system of claim 10, wherein the plurality of instructions, when executed by the processor, cause the processor to cause a display, by the display device, of an indication of different probabilities of winning based on how many rows of symbol display positions are selected as active rows of symbol display positions.

12. A gaming system comprising:

a processor; and

a memory device that stores a plurality of instructions that, when executed by the processor, cause the processor to:

cause a display, by a display device, of symbol displays comprising different symbols;

cause a display, by the display device, of an indication of selectable symbol display positions associated with the symbol display;

cause the processor to cause a display, by the display device, of an indication of different probabilities of winning based on how many symbol display positions are selected as active symbol display positions;

for a first play of a game:

cause a display, by the display device, of an indication of a first quantity of the symbol display positions as being first active symbol display positions for the first play of the game;

cause a display, by the display device, of a first plurality of the different symbols on the symbol displays displayed at the first active symbol display positions; and

cause a display, by the display device, of an indication of a first quantity of same symbols at the first active symbol display positions, wherein whether the first quantity of same symbols is a winning symbol combination depends on the symbol display positions selected as the first active symbol display positions, wherein the first quantity of same symbols at the first active symbol display positions does not form the winning symbol combination; and

for a second play of the game:

cause a display, by the display device, of an indication of a second quantity of the symbol display positions as being second active symbol display positions for the second play of the game;

cause a display, by the display device, of a second plurality of the different symbols on the symbol displays displayed at the second active symbol display positions; and

cause a display, by the display device, of an indication the first quantity of same symbols at the second active symbol display positions, wherein whether the first quantity of same symbols is the winning symbol combination depends on the symbol display positions selected as the second active symbol display positions, wherein the first quantity of same symbols at the second active symbol display positions form the winning symbol combination.

13. The gaming system of claim 12, wherein the display, by the display device, of the indication of selectable symbol display positions comprises an indication of different selectable columns of the symbol display positions as active columns of symbol display positions.

14. The gaming system of claim 13, wherein the plurality of instructions, when executed by the processor, cause the processor to cause a display, by the display device, of an indication of different probabilities of winning based on how many columns of symbol display positions are selected as 5 active columns of symbol display positions.

15. The gaming system of claim 12, wherein the display, by the display device, of the indication of selectable symbol display positions comprises an indication of different selectable rows of the symbol display positions as active rows of 10 symbol display positions.

16. The gaming system of claim 12, wherein the plurality of instructions, when executed by the processor, cause the processor to cause a display, by the display device, of an indication of different probabilities of winning based on how 15 many rows of symbol display positions are selected as active rows of symbol display positions.

17. The gaming system of claim 12, wherein the display, by the display device, of the indication of selectable symbol display positions comprises an indication of different selectable groups of the symbol display positions as active groups 20 of symbol display positions.

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